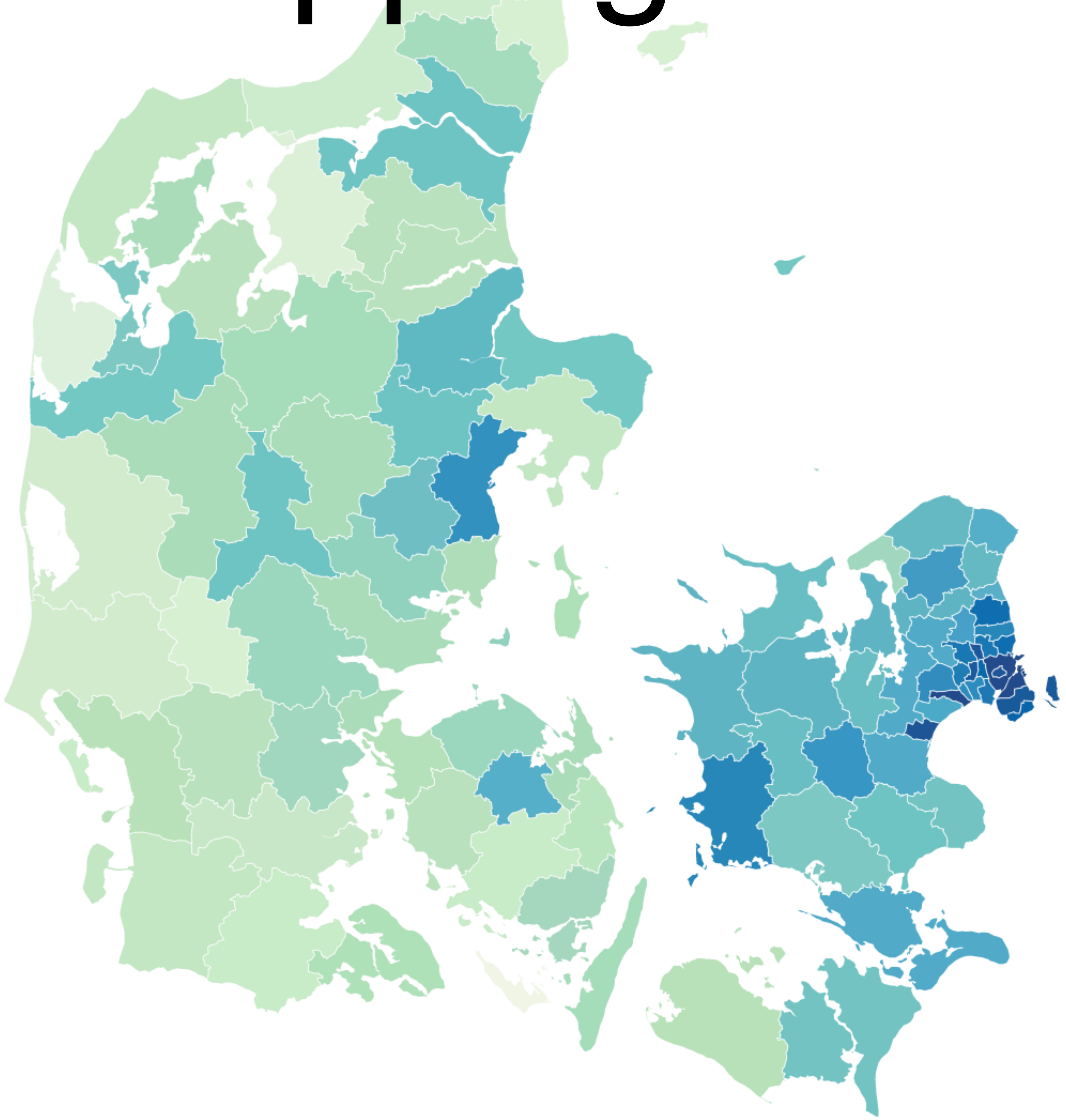
Lecture 3: Choropleth mapping

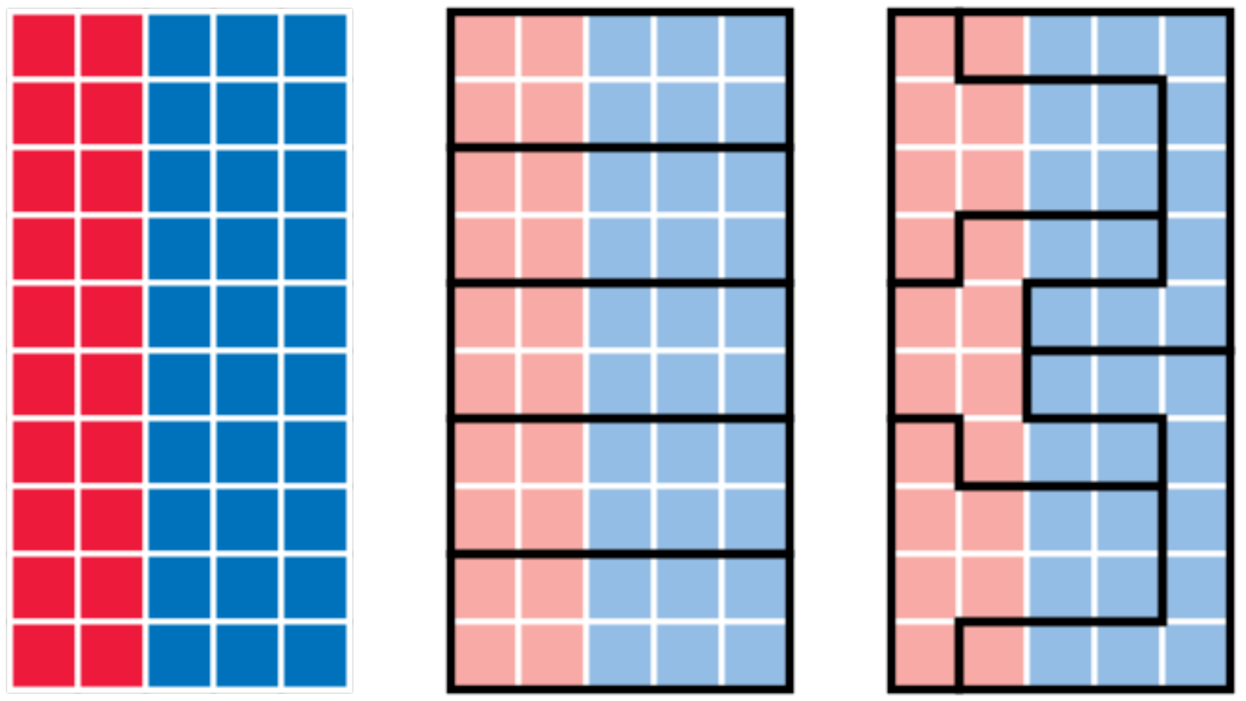
Instructor: Michael Szell
Feb 11, 2026



IT UNIVERSITY OF COPENHAGEN

Today you will learn about MAUP and Choropleths

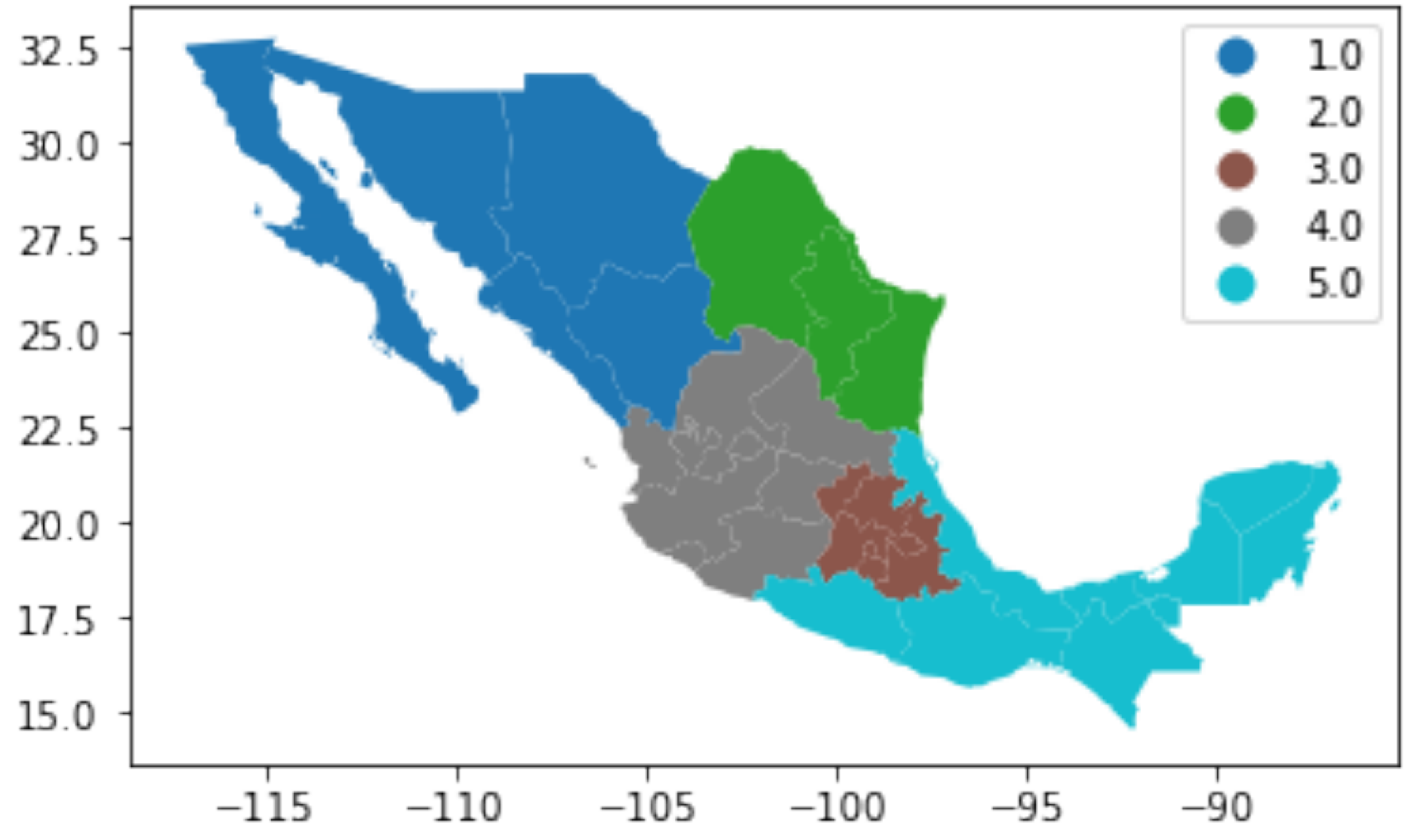
Biases from spatial aggregation (MAUP)



Classification schemes

$$c_j < y_i \leq c_{j+1} \; \forall y_i \in C_j$$

Choropleths in Python



The most common steps in Data Preprocessing are:

- Aggregation
- Sampling
- Dimensionality reduction
- Discretization / Classification
- Variable transformation

Aggregation = Combining objects into a single one

Student ID	Year	Grade Point Average (GPA)	...
	⋮		
1034262	Senior	3.24	...
1052663	Sophomore	3.51	...
1082246	Freshman	3.62	...
	⋮		

NULL	Non-Freshman	3.375

Aggregation = Combining objects into a single one

Examples:

GPS coordinate → Zip Code → City → Country

Second → Minute → Hour → Day → Week → Month → Year

Advantages: Data reduction, easier to process, high-level view, smaller statistical fluctuations

Disadvantages: Loss of details, introducing biases

Choropleth maps visualize geospatial data with colors

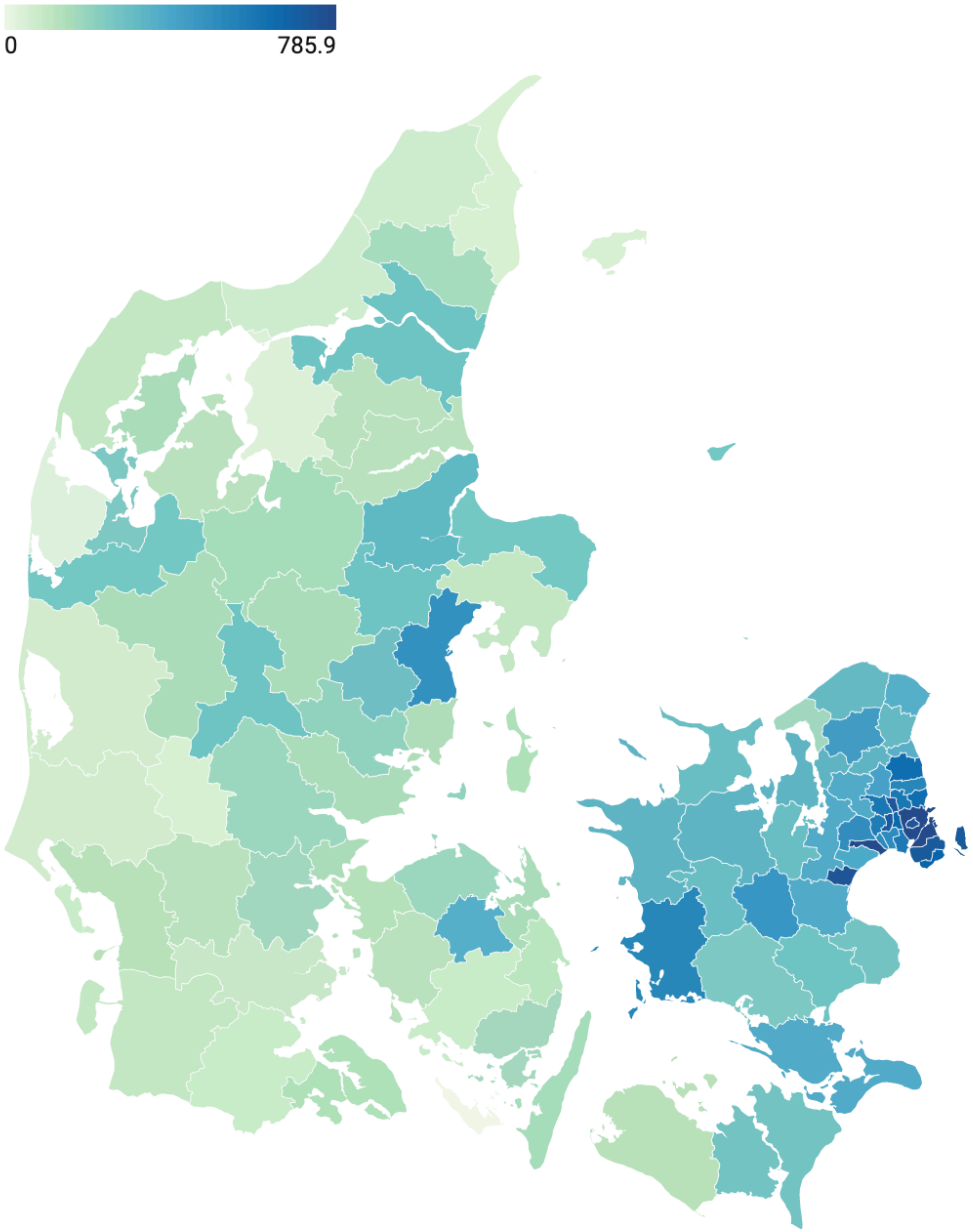
Choro = region
pleth = multiple

Combine: Polygons + Attribute values

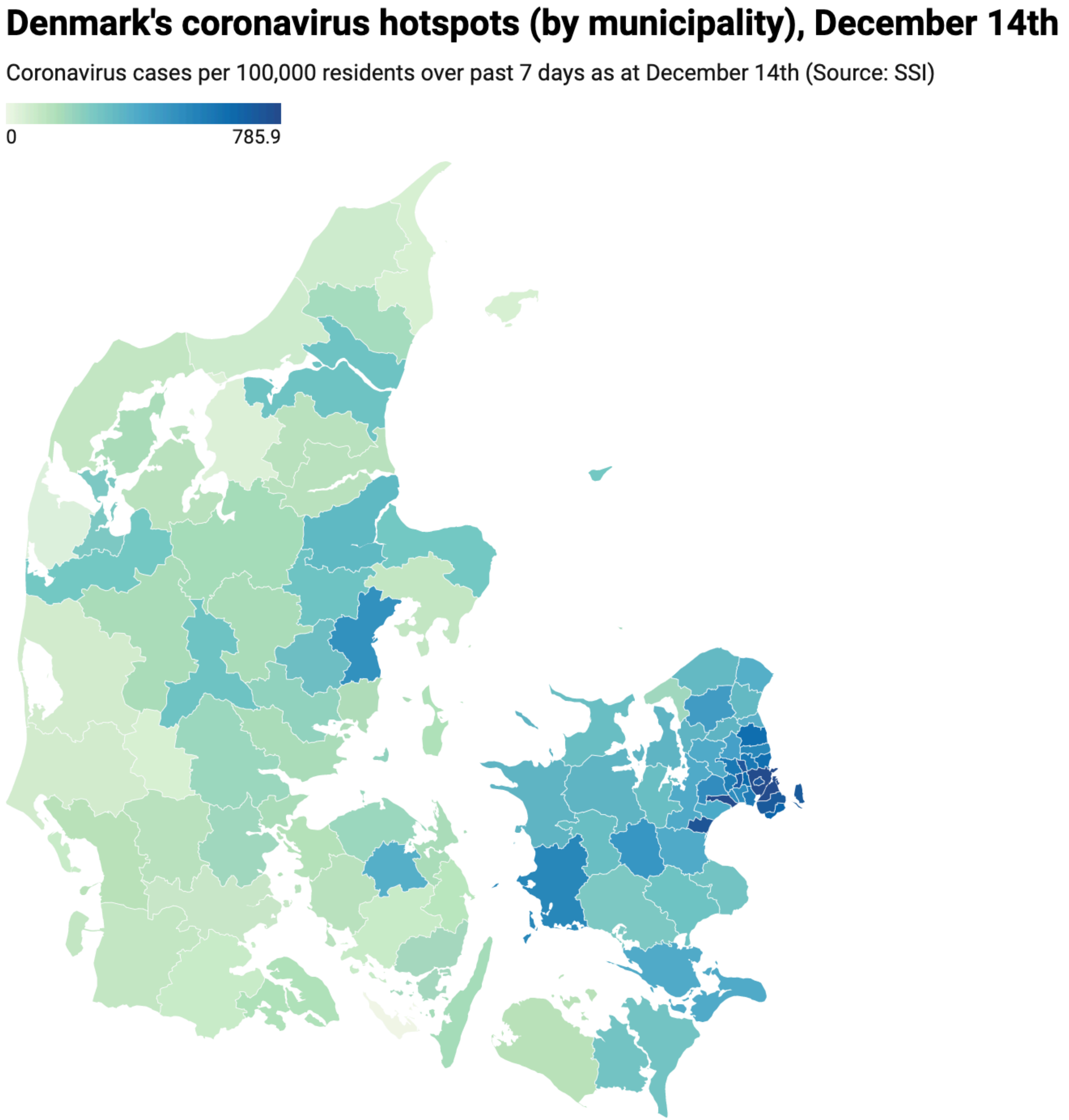
Values are often grouped into classes to reduce complexity

Denmark's coronavirus hotspots (by municipality), December 14th

Coronavirus cases per 100,000 residents over past 7 days as at December 14th (Source: SSI)



https://www.thelocal.dk/20201028/maps-where-are-denmarks-coronavirus-hotspots/



3 choices:

1) spatial units

2) colors

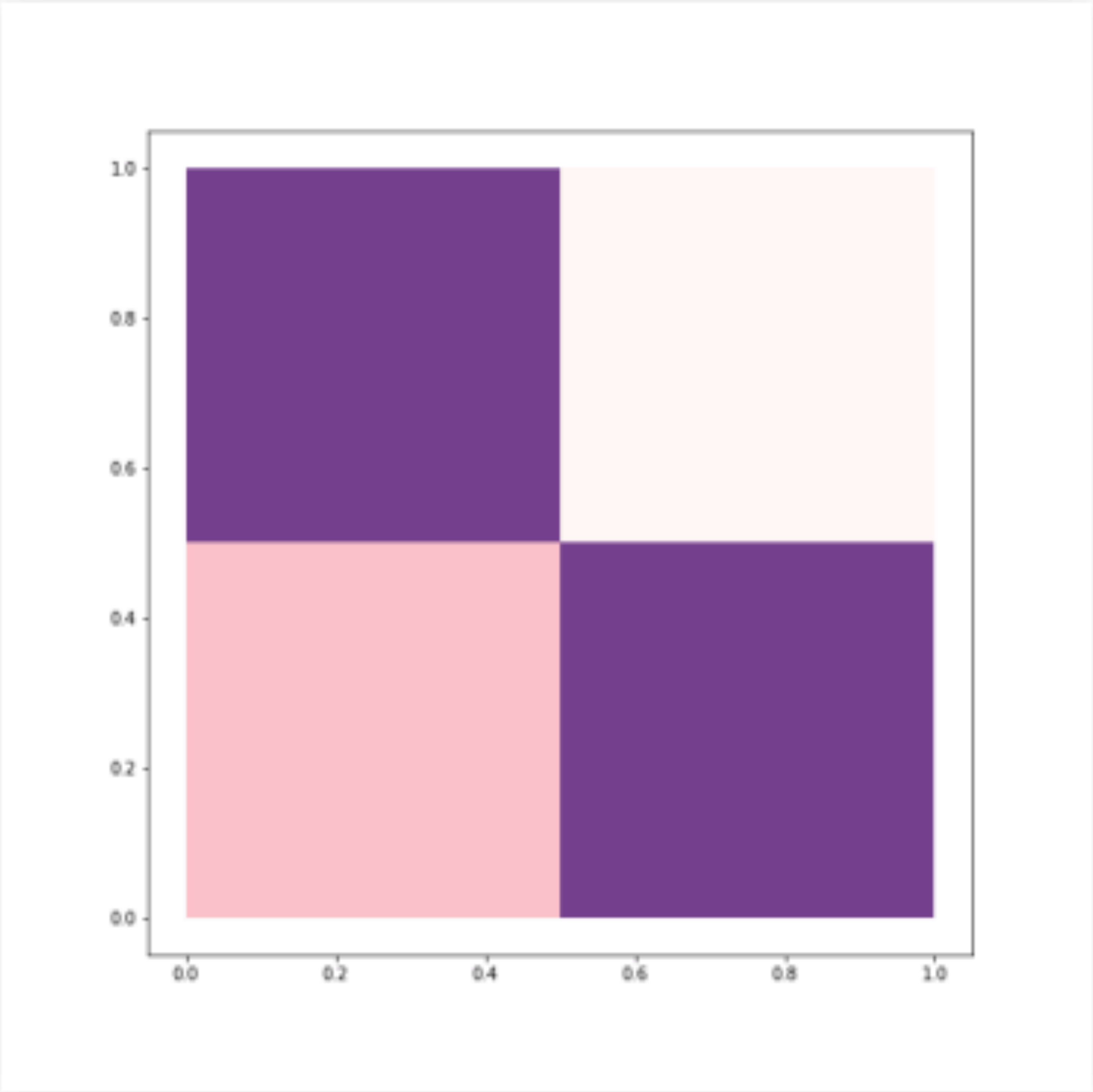
3) classes

https://www.thelocal.dk/20201028/maps-where-are-denmarks-coronavirus-hotspots/

Population density (people aggregated in space)

Group experiment

Each of 3 groups describes their picture in 1 sentence


1.0
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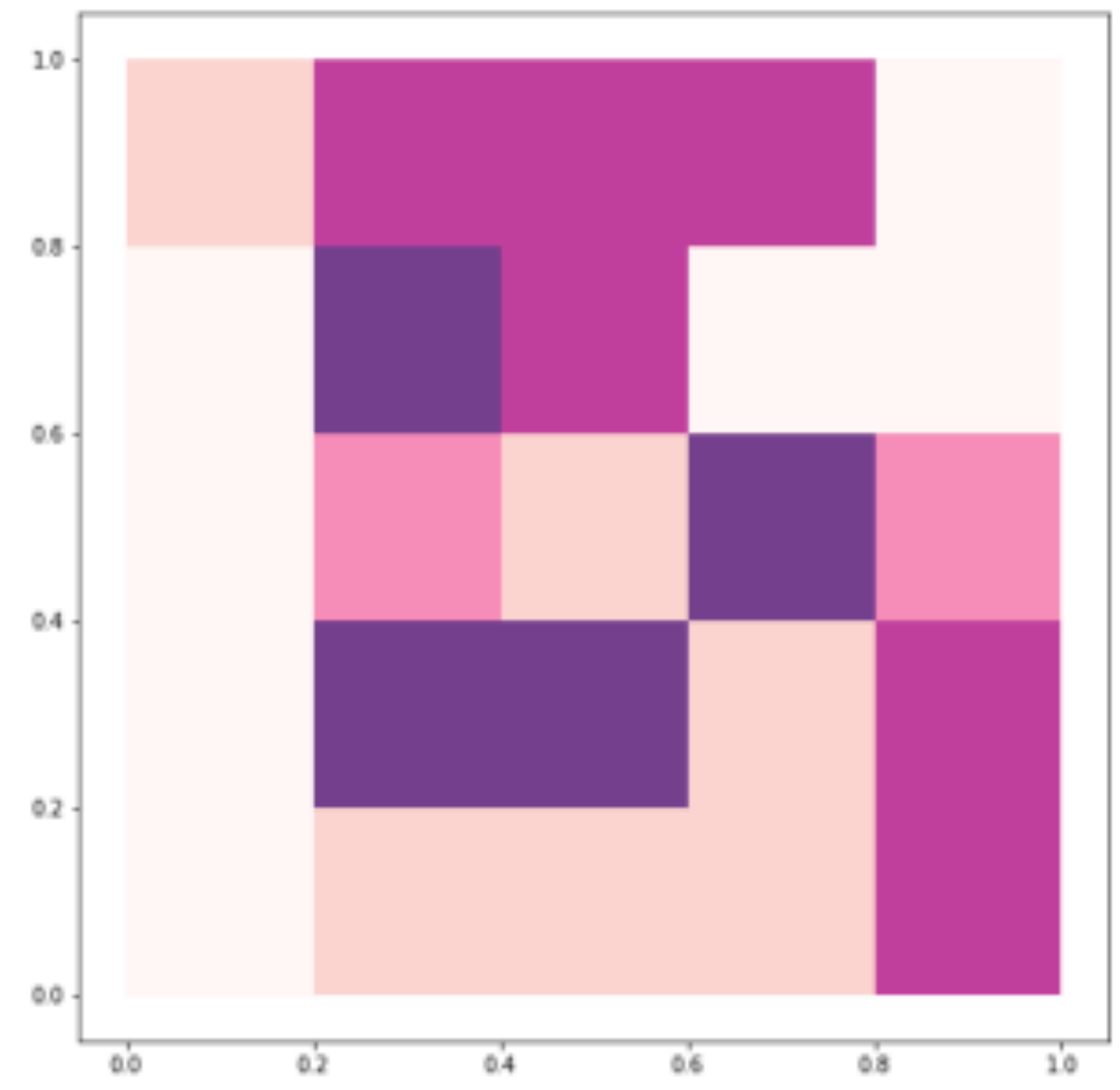

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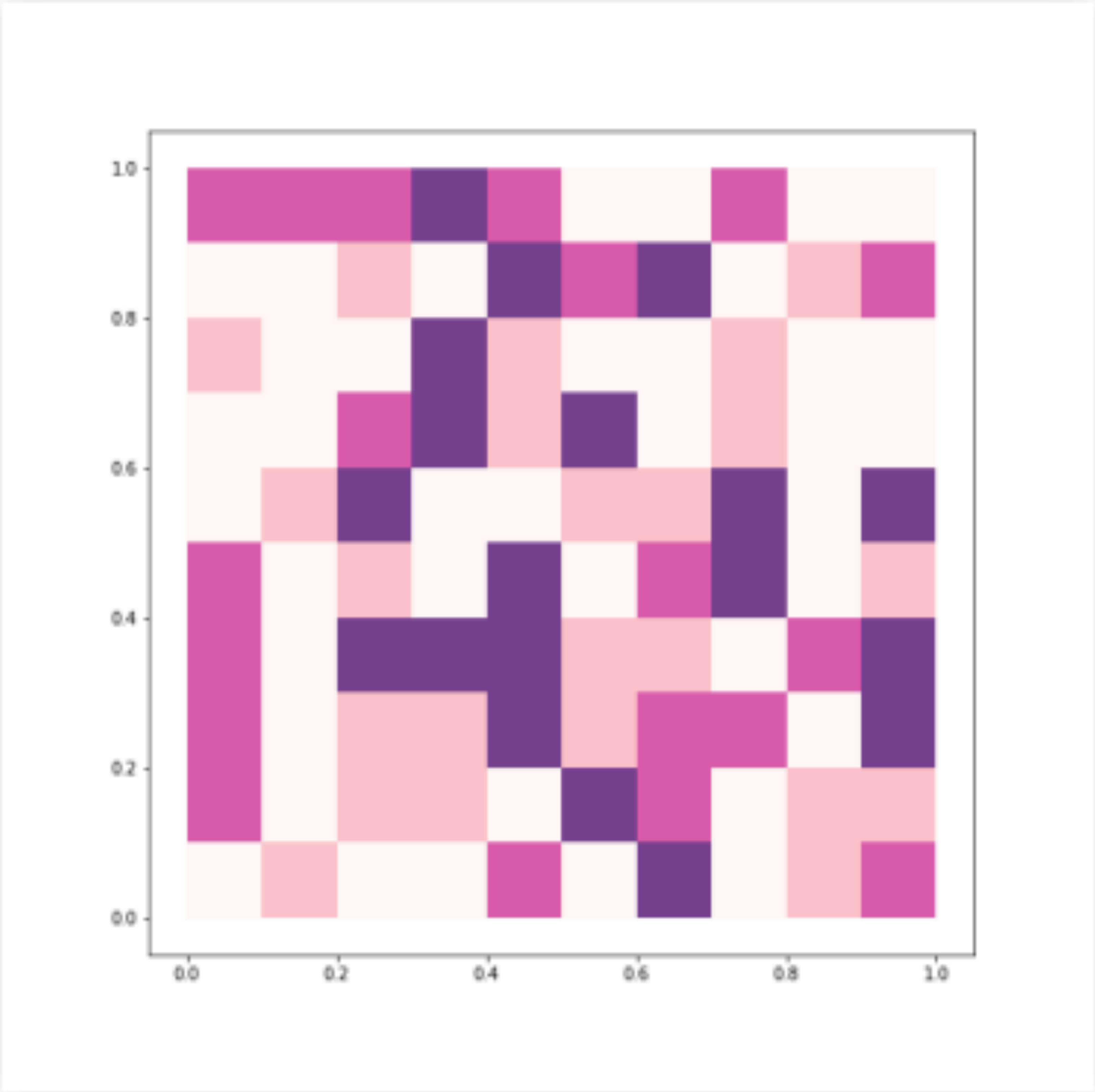

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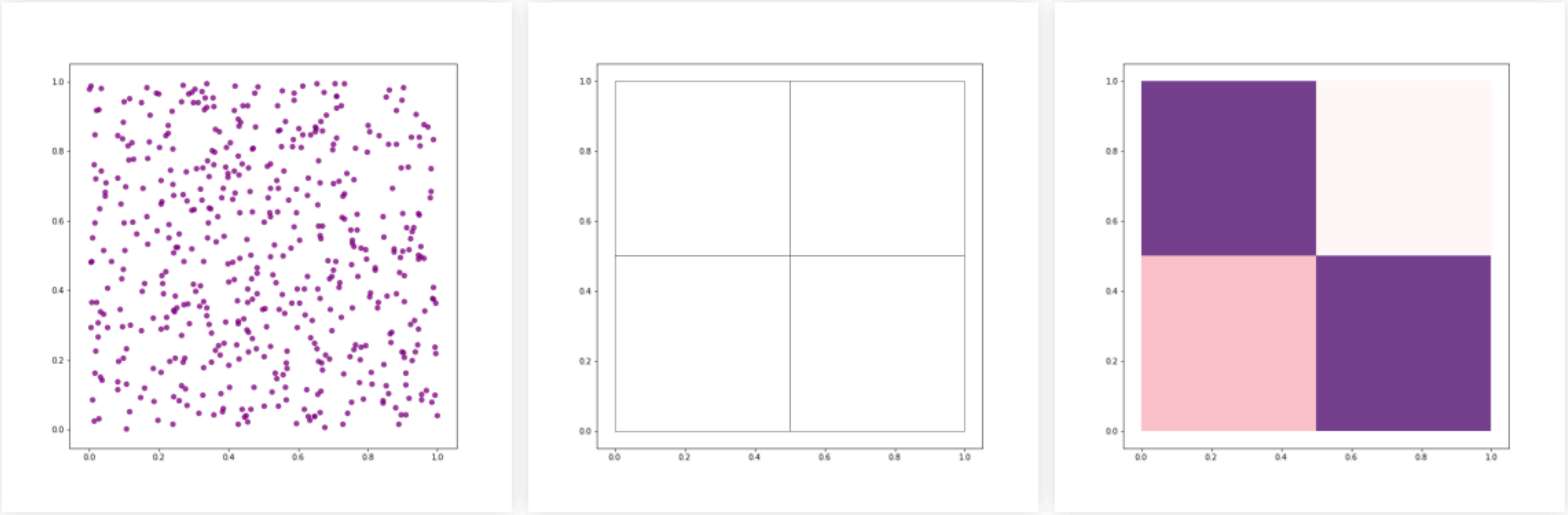
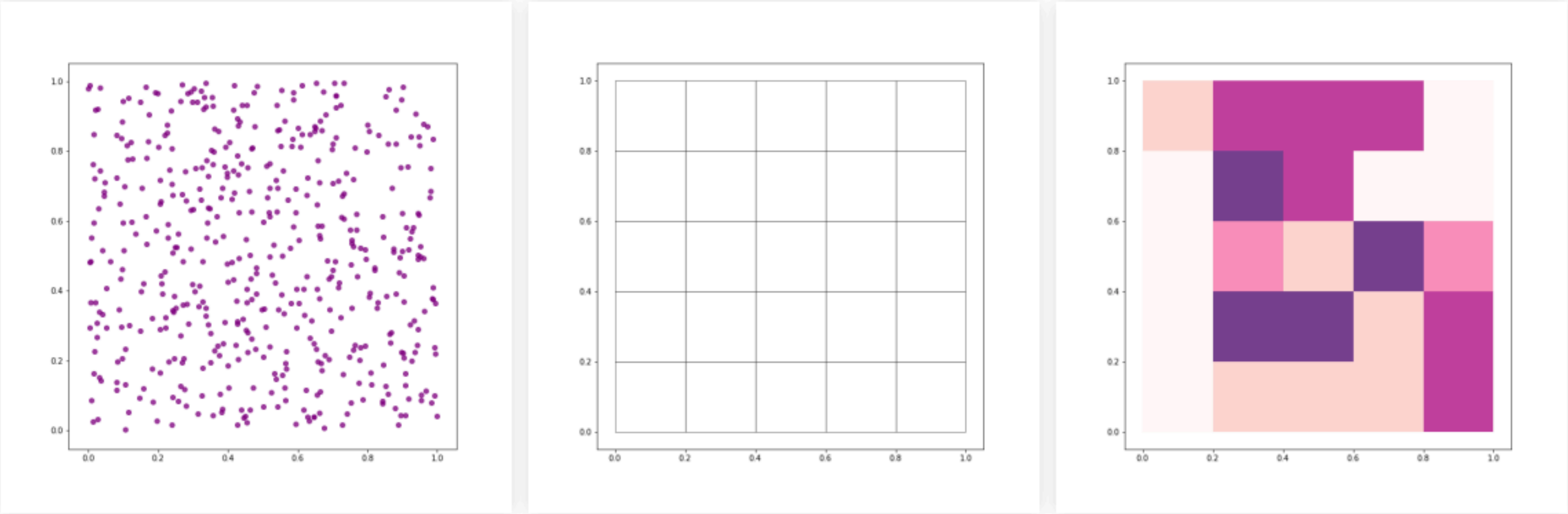


https://gist.github.com/darribas/8b5a7b93d4085223f1c5



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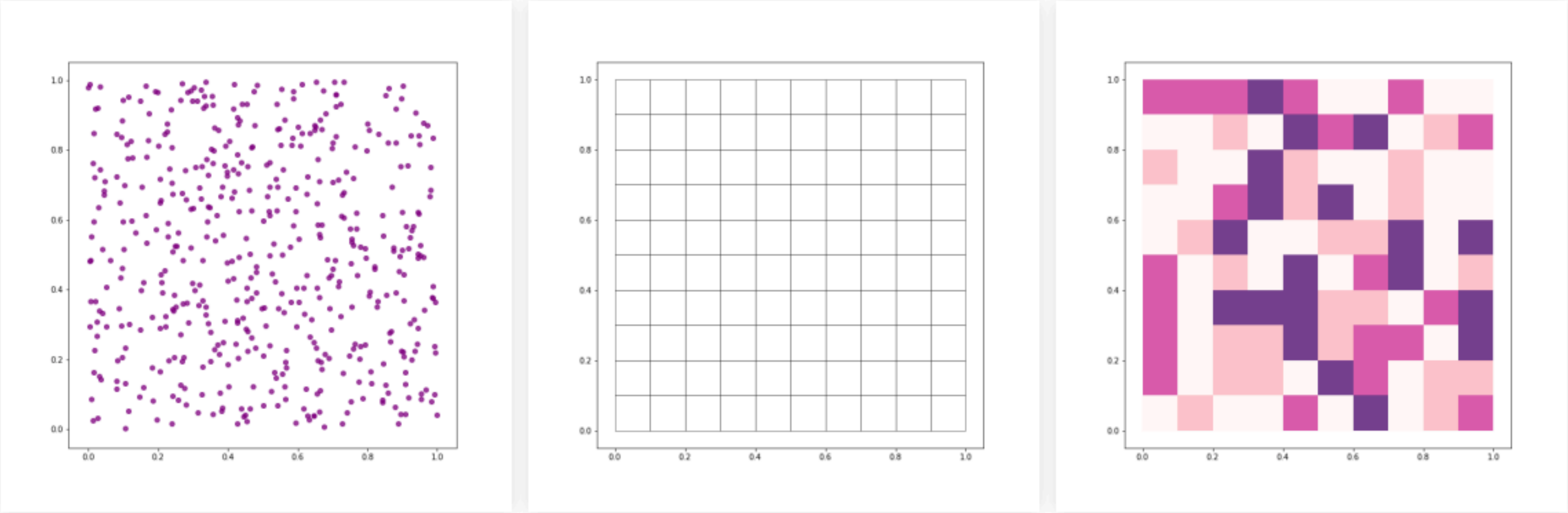
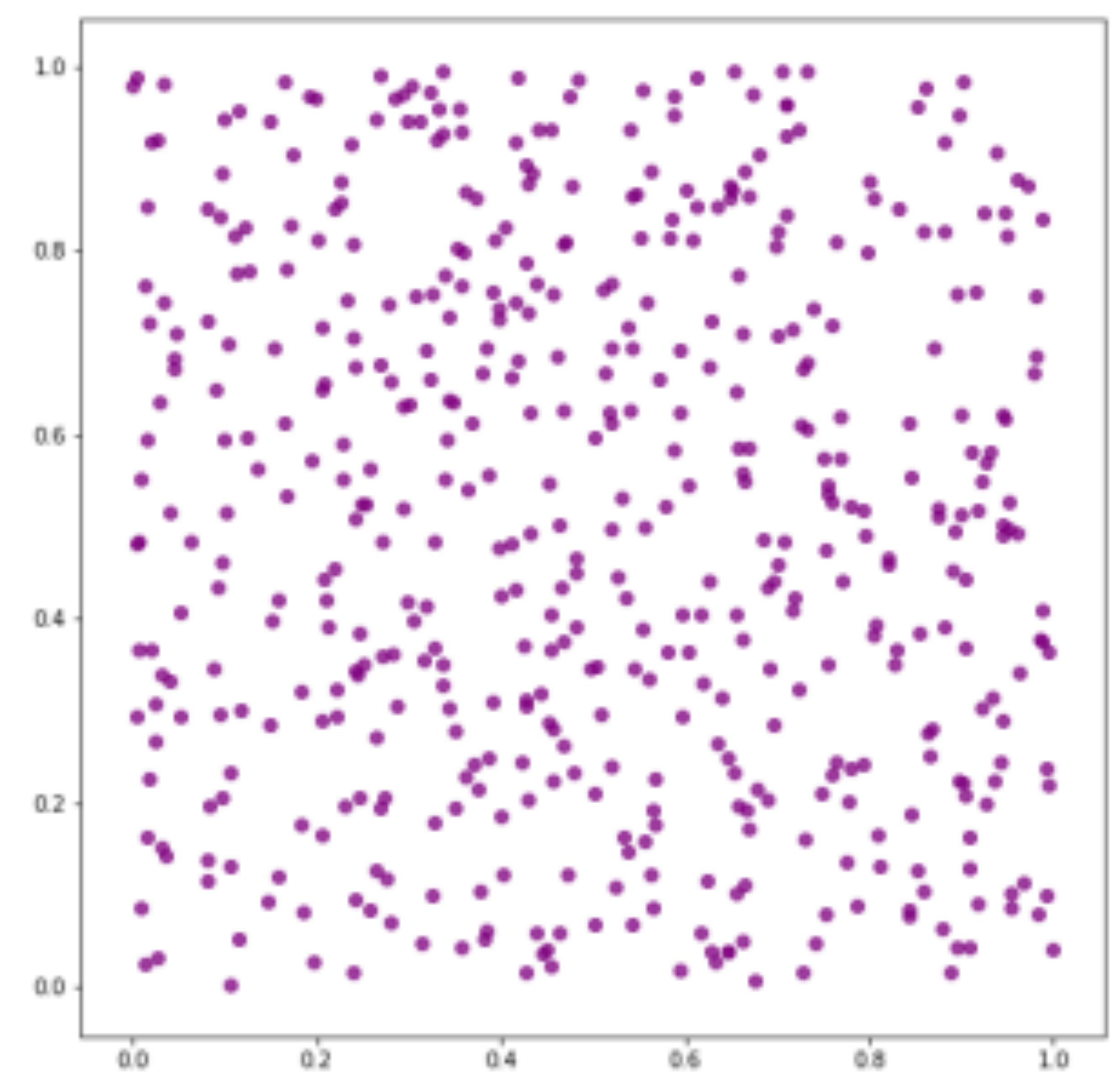
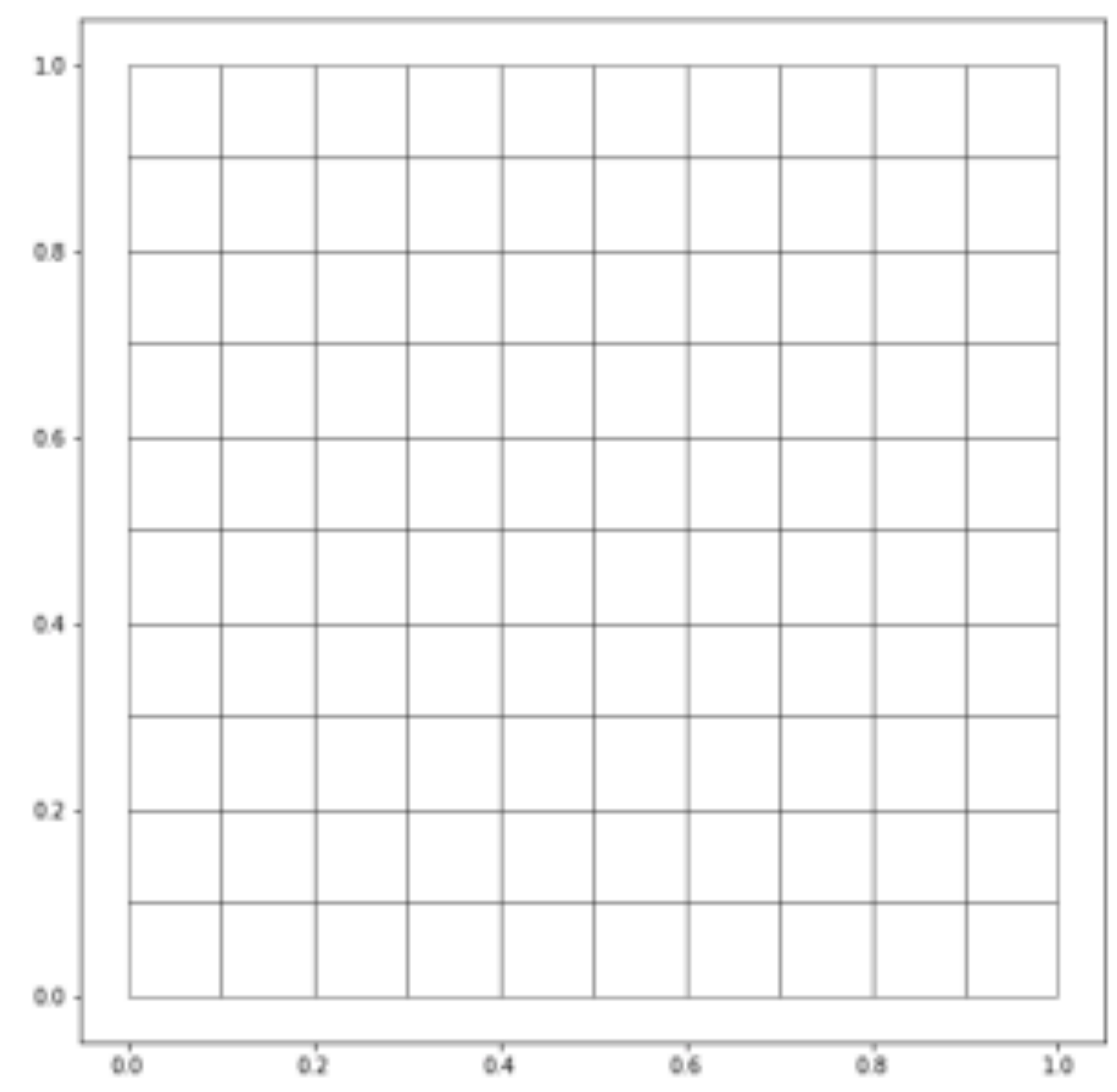


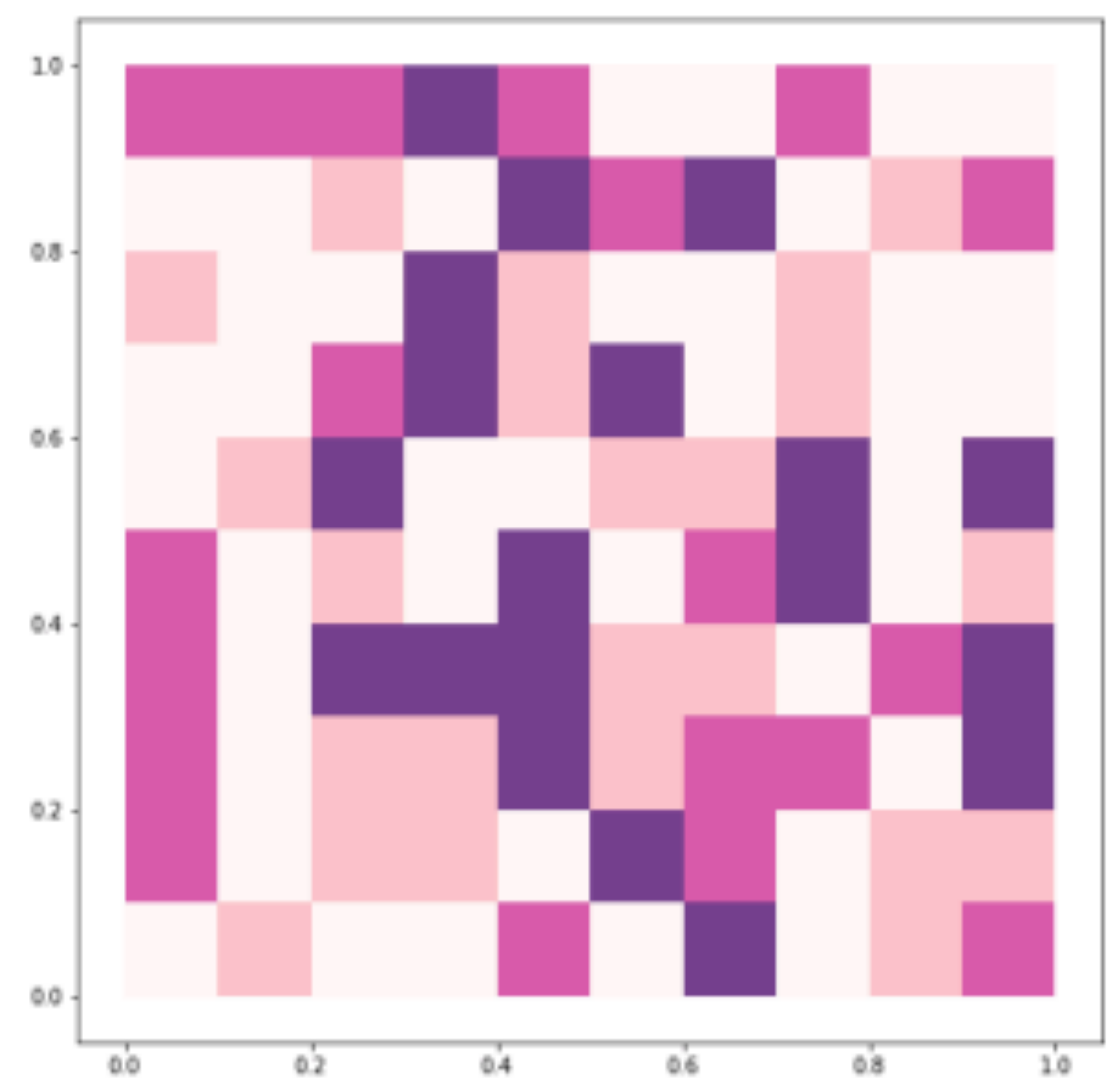
https://gist.github.com/darribas/8b5a7b93d4085223f1c5

Same RANDOM data

Same colors
Same classification



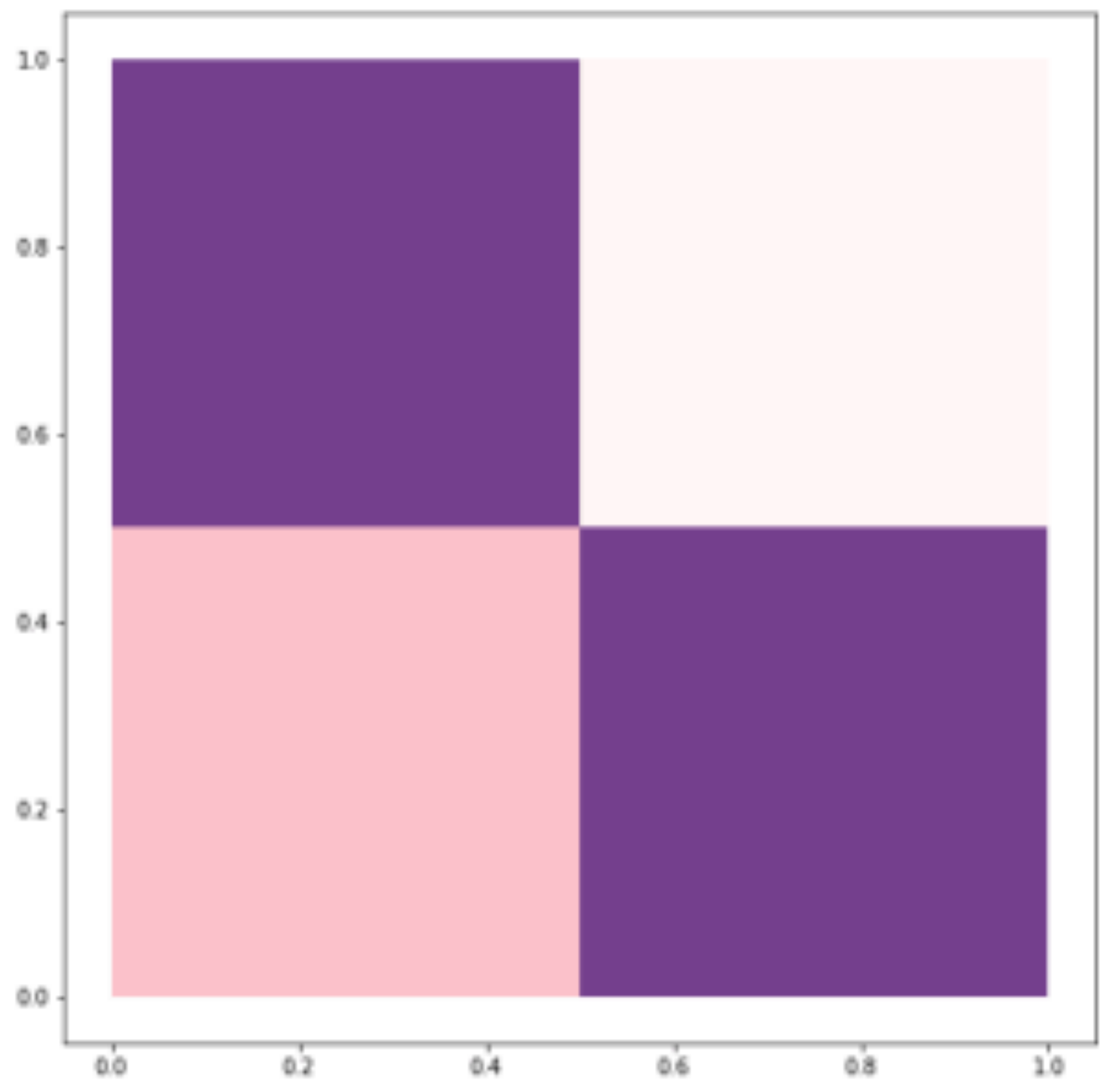


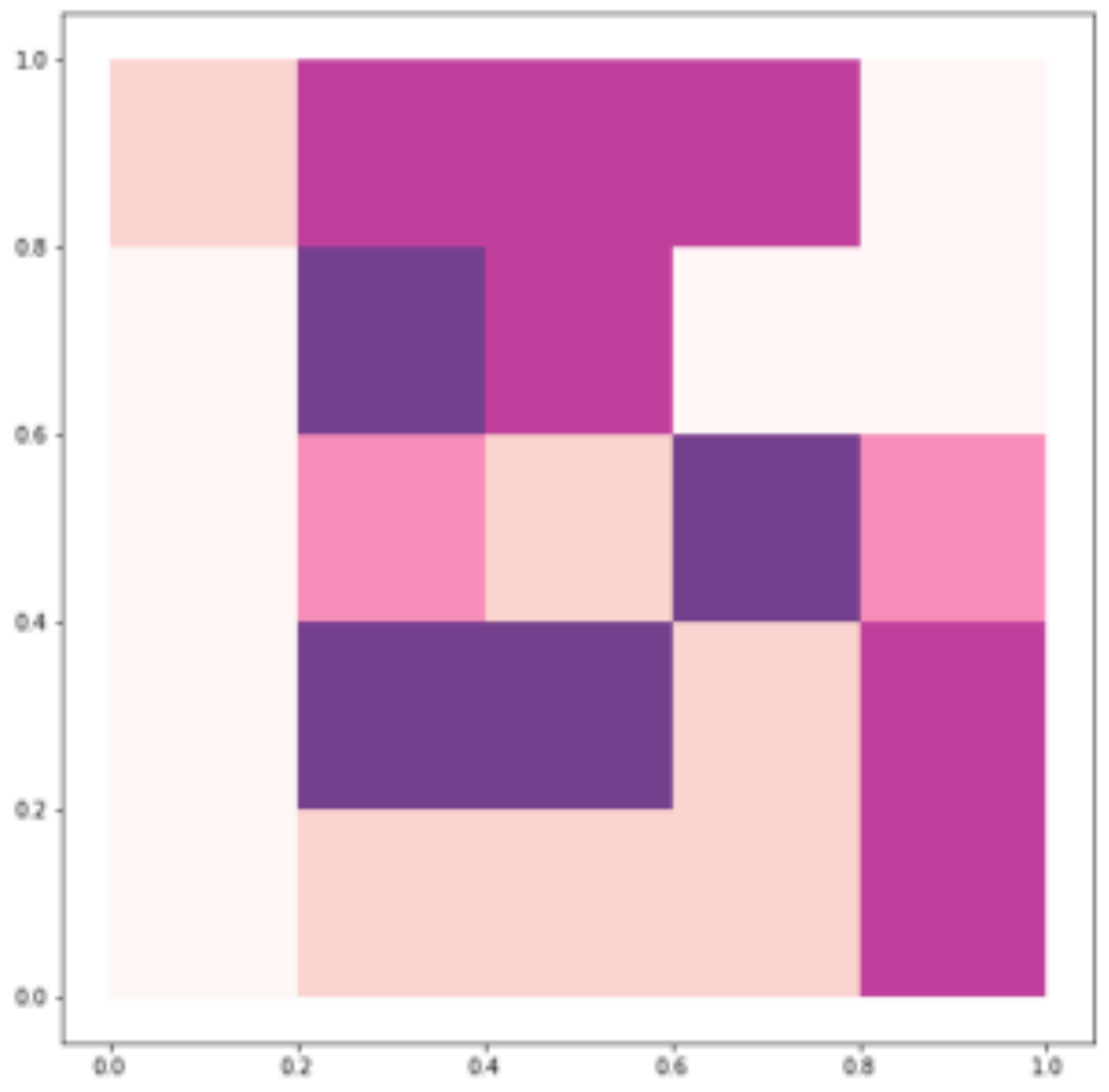


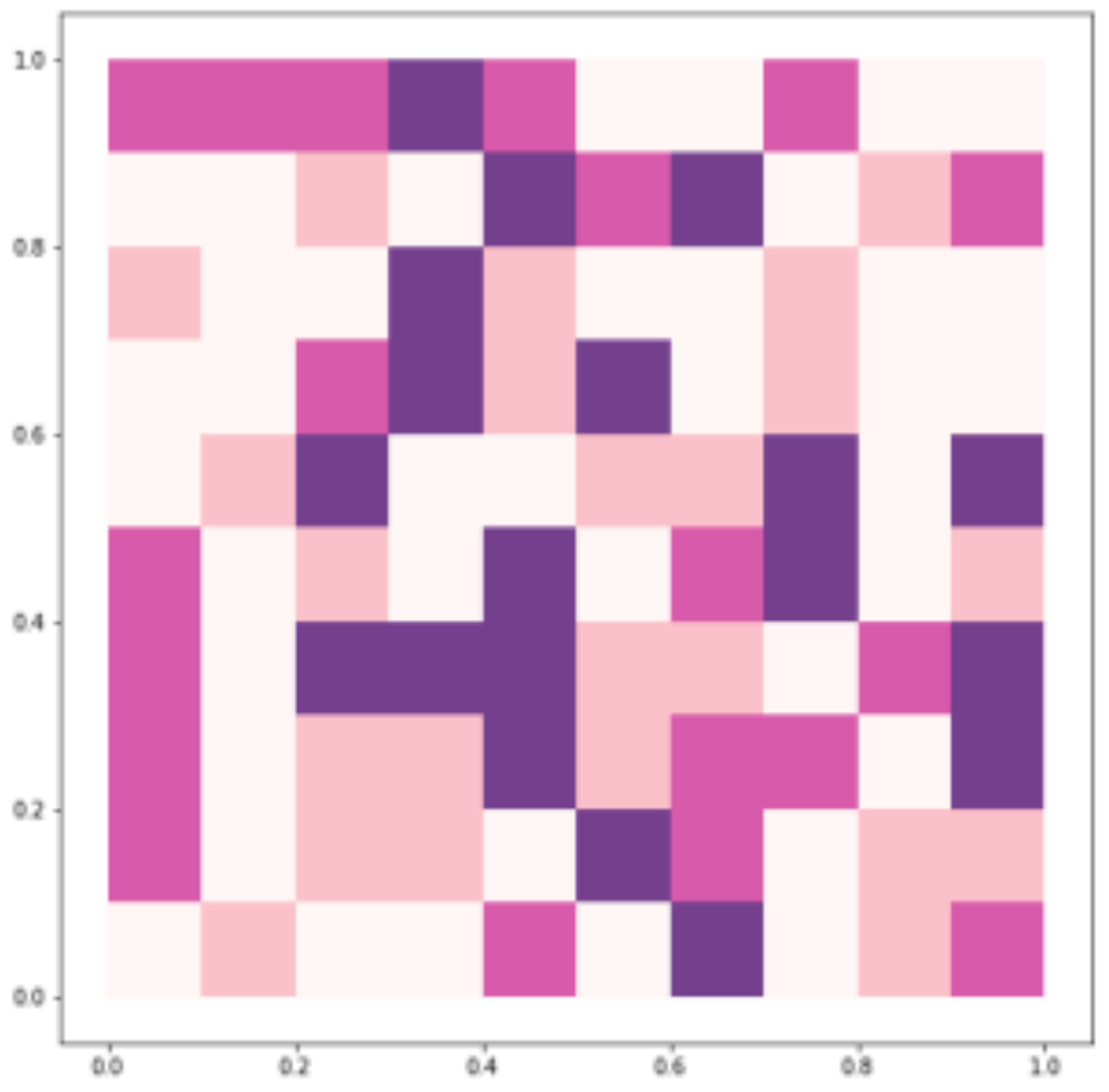
Different spatial units

https://gist.github.com/darribas/8b5a7b93d4085223f1c5

A common source of bias in spatial aggregation: MAUP





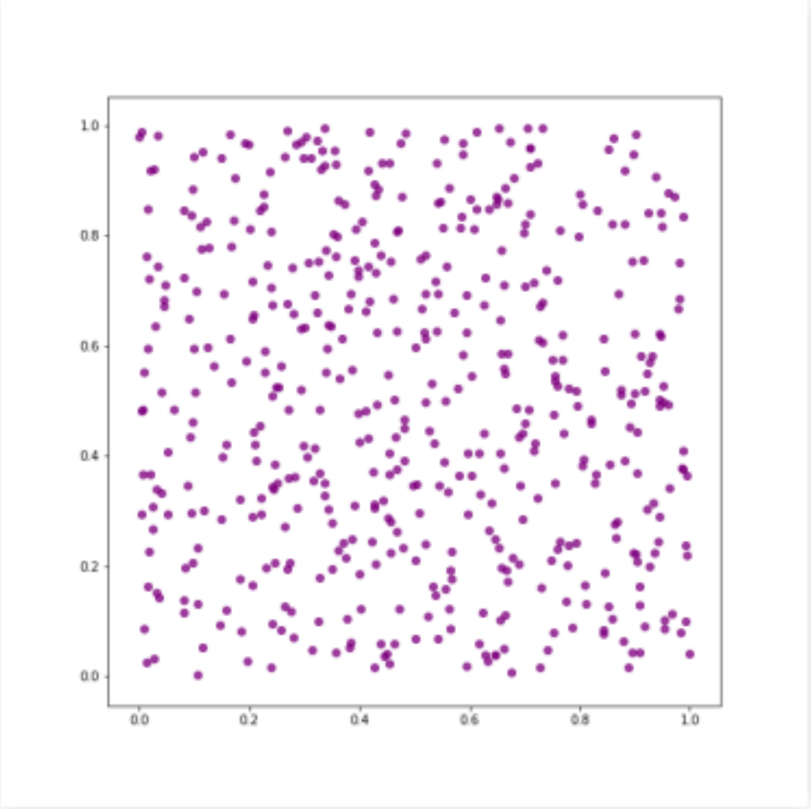


https://gist.github.com/darribas/8b5a7b93d4085223f1c5

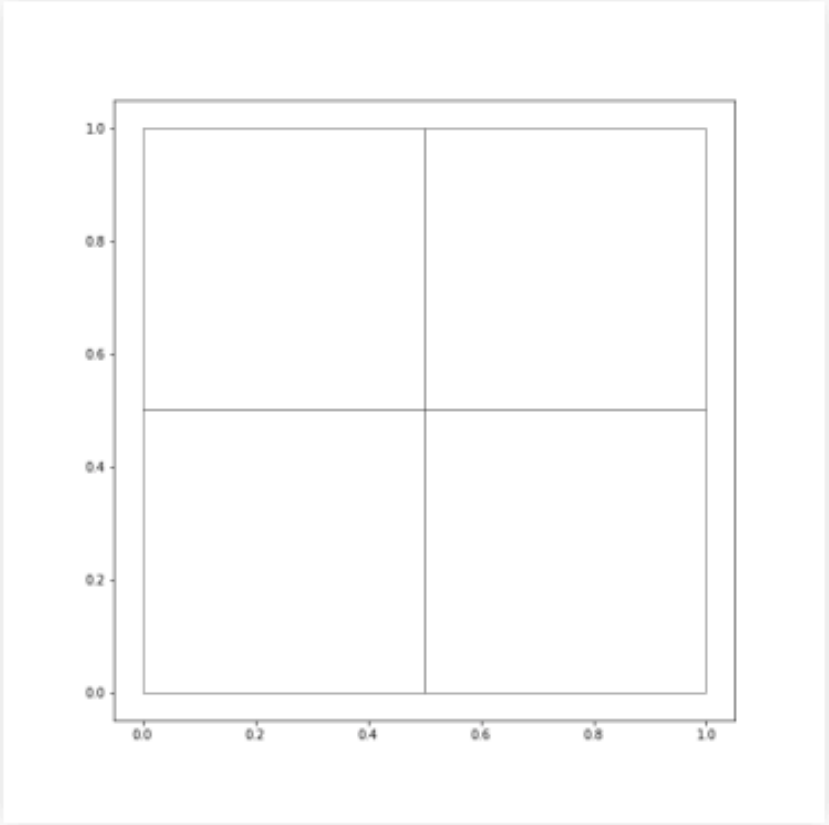
A common source of bias in spatial aggregation: MAUP

The MAUP (Modifiable Areal Unit Problem) is a scale and delineation mismatch between:

Underlying entities $\longleftrightarrow$ Unit of measurement



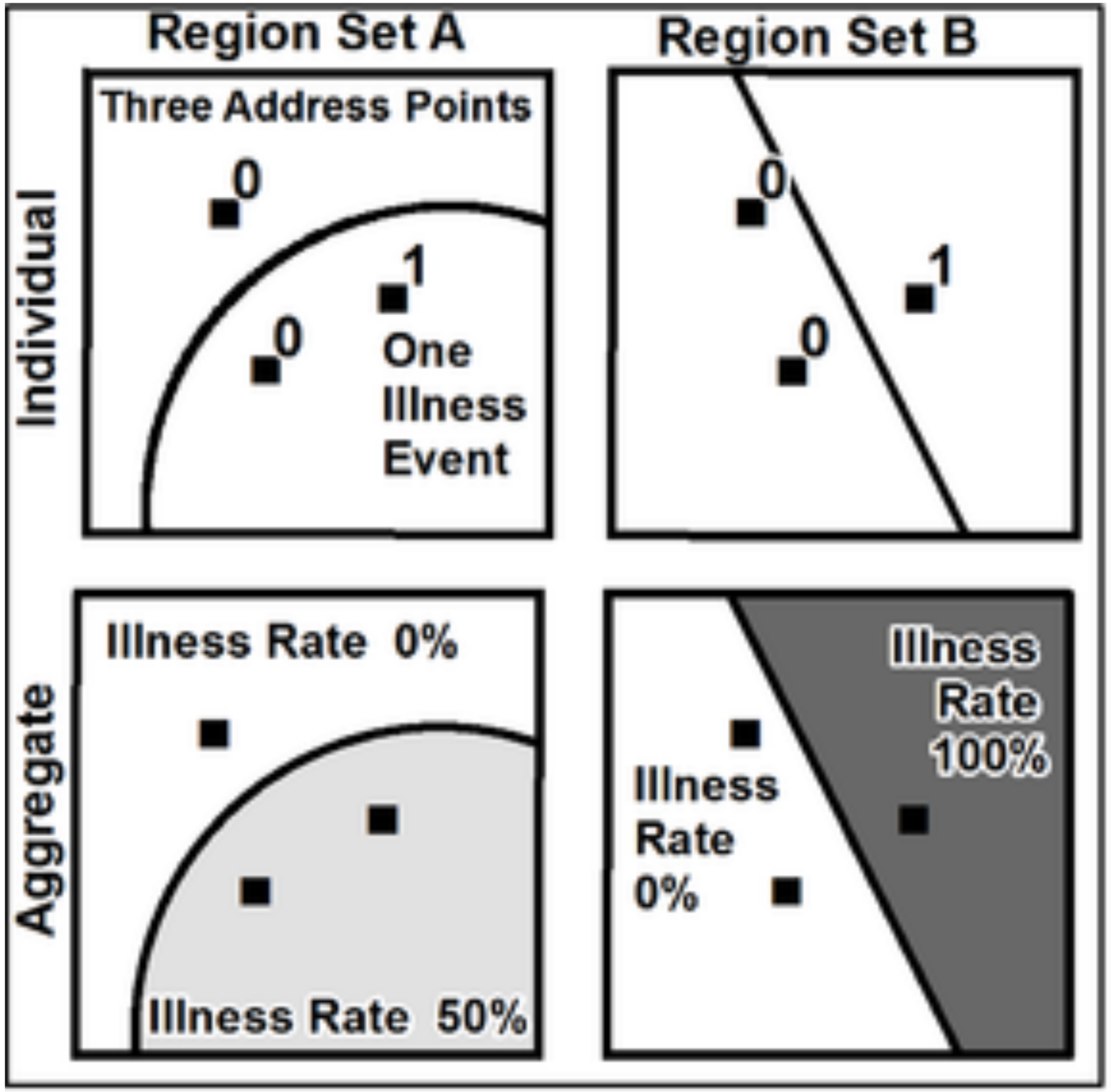
Individuals, shops,..



Districts, regions, resolution,..

Openshaw, 1984

A common source of bias in spatial aggregation: MAUP



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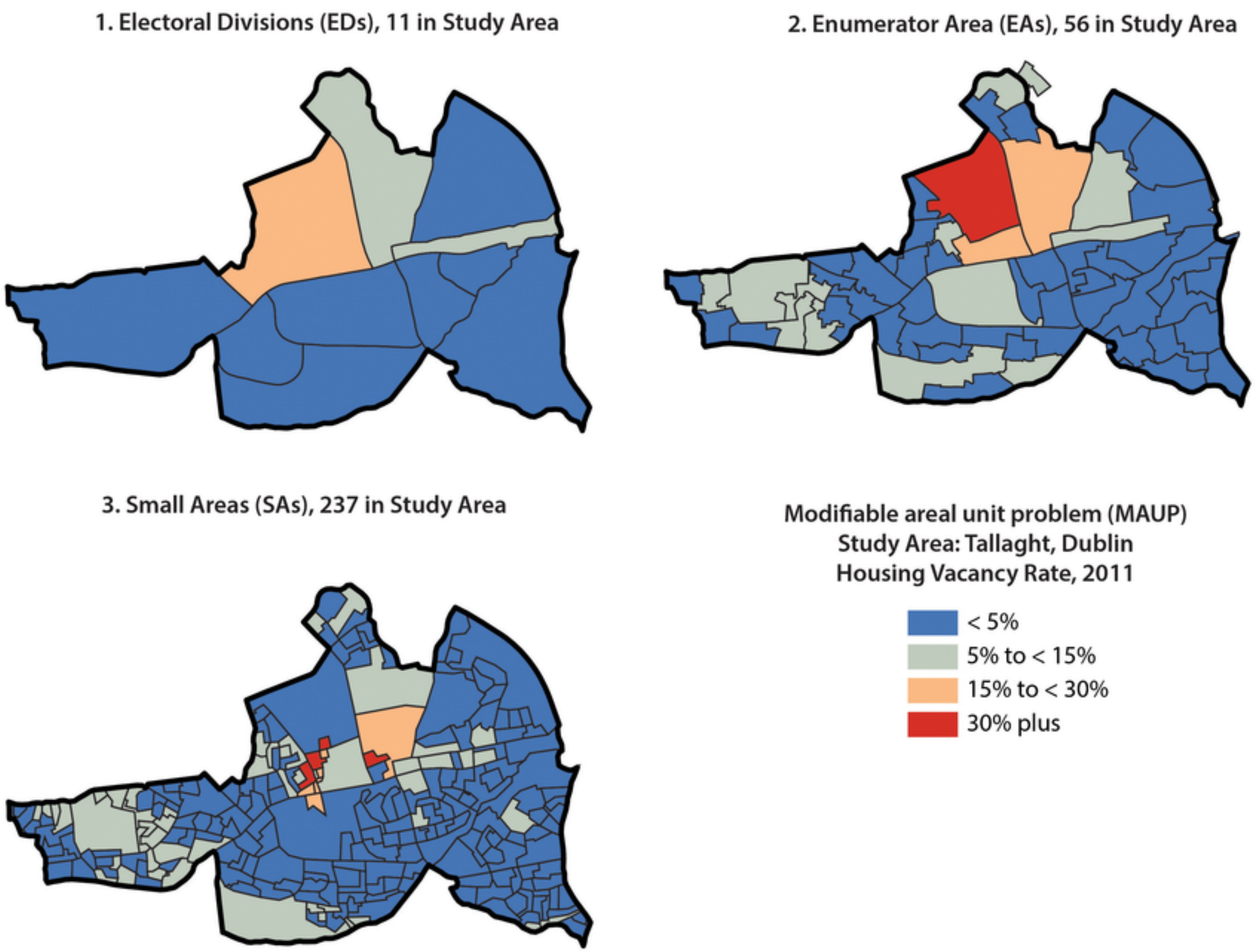
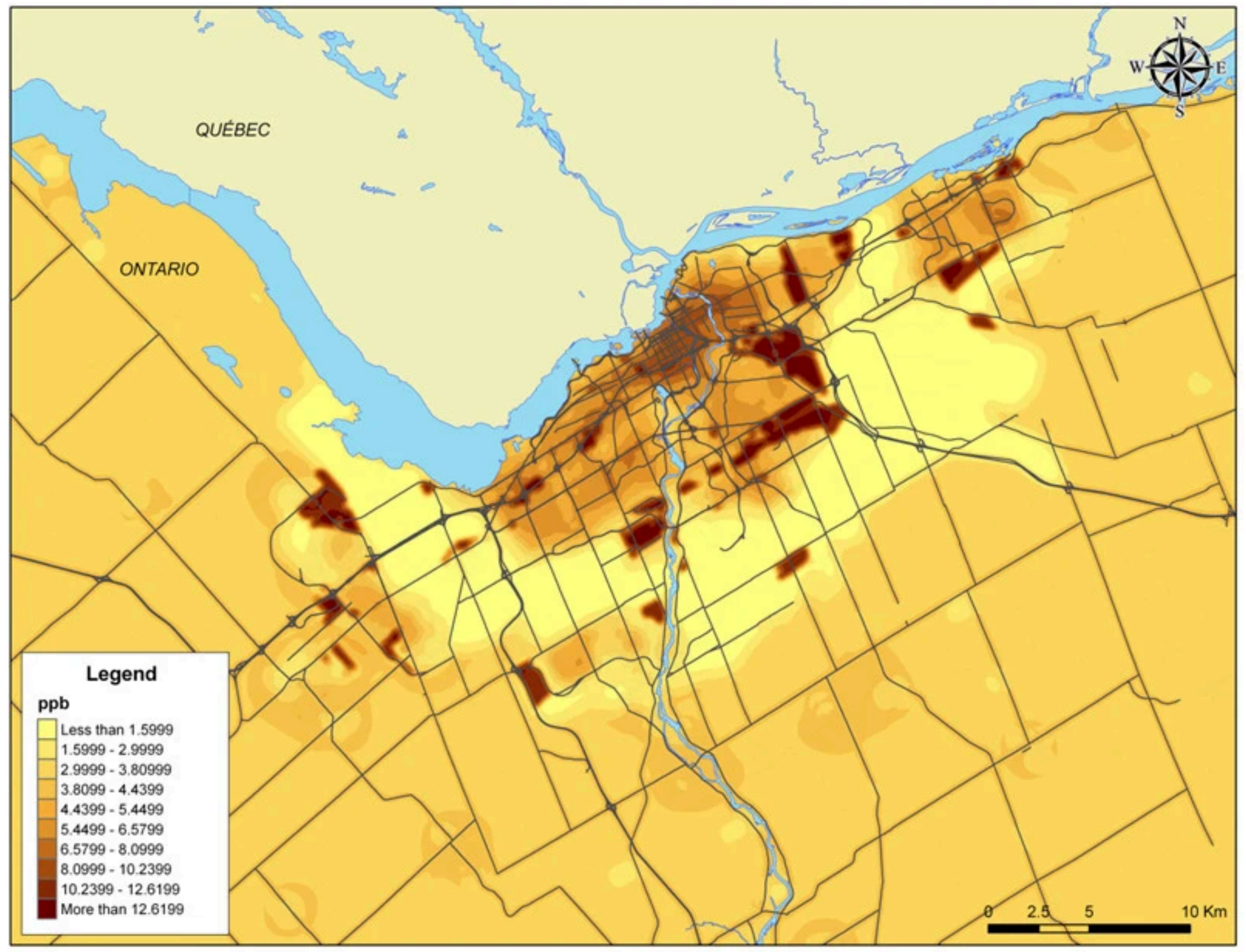


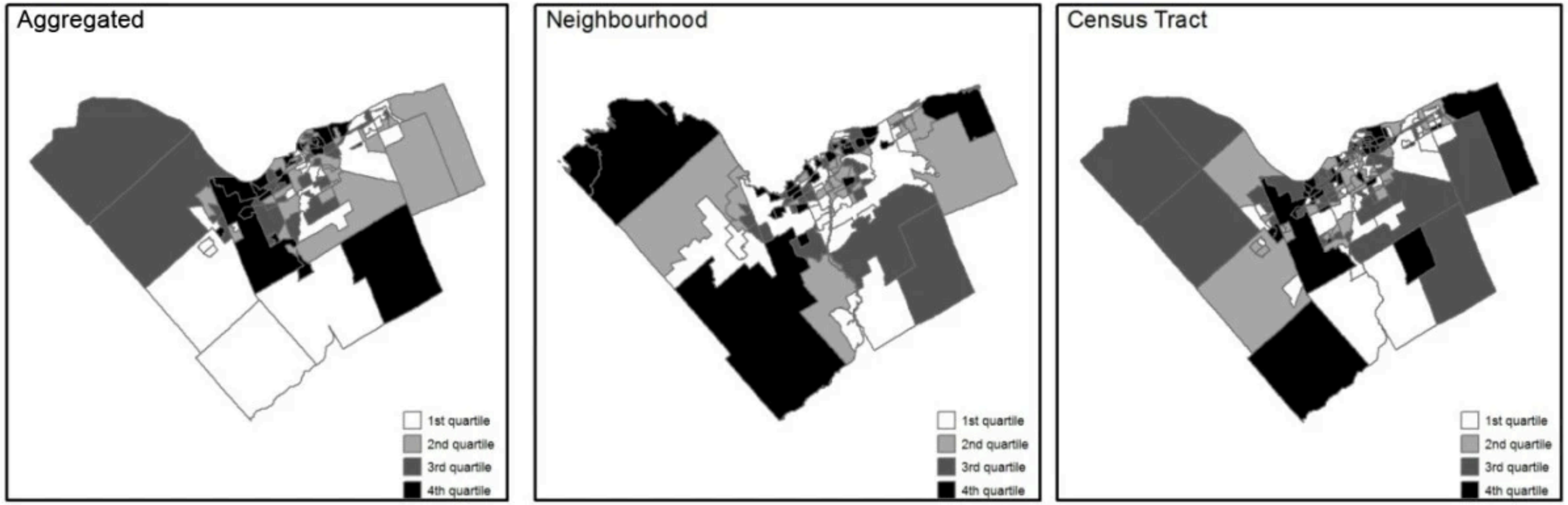
https://www.tandfonline.com/doi/full/10.1080/21681376.2014.983149

A common source of bias in spatial aggregation: MAUP



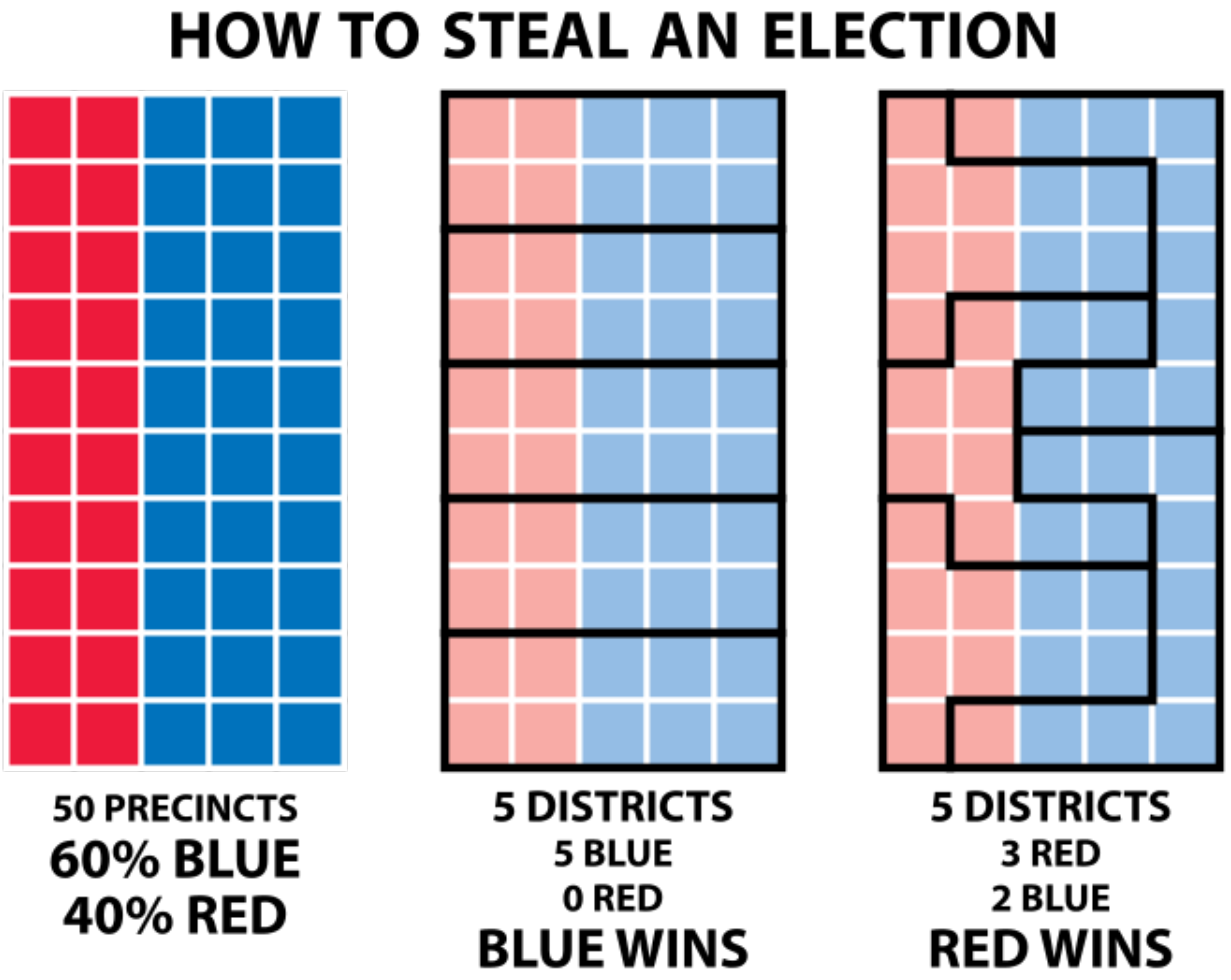
Pollution

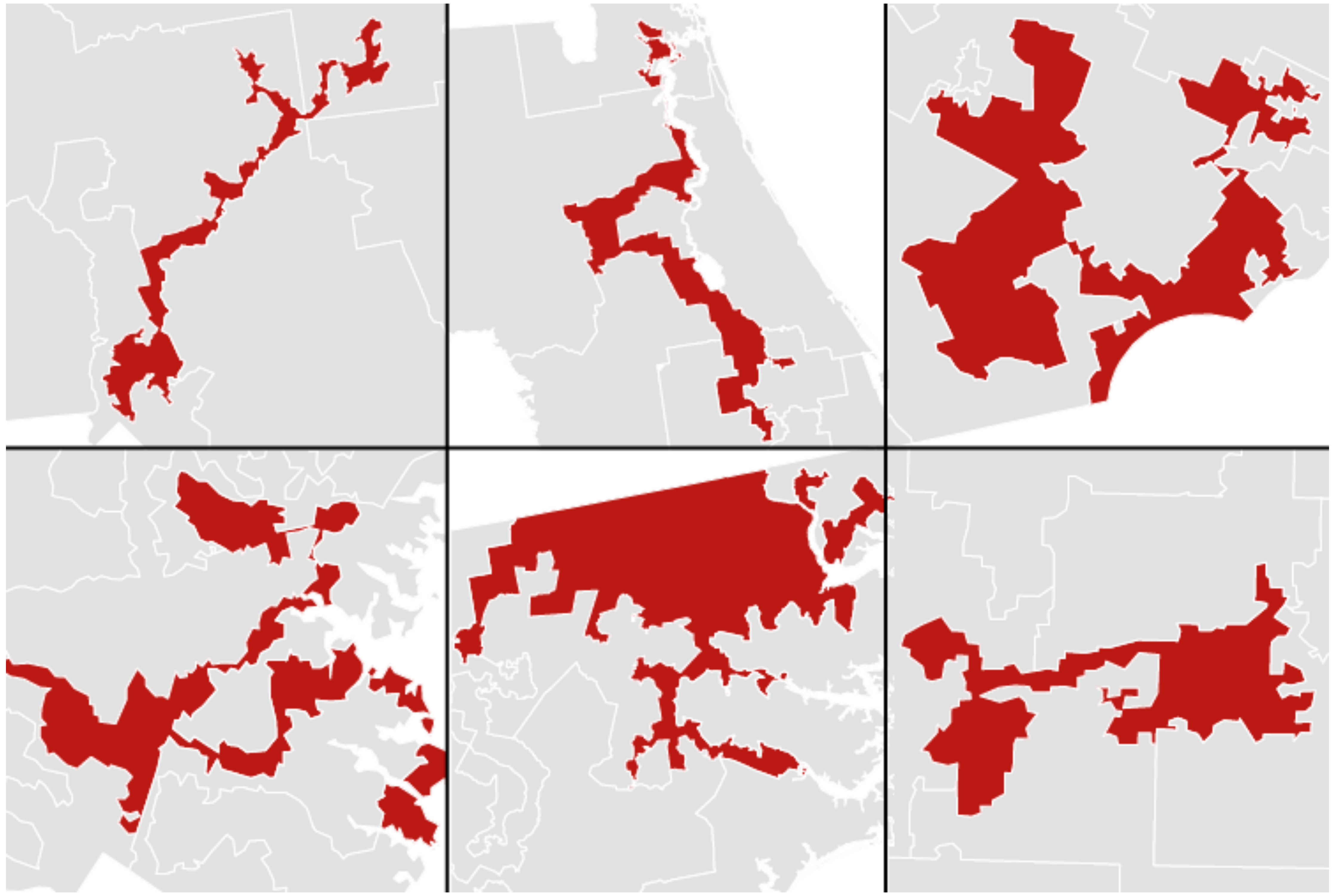
Respiratory Morbidity



https://ij-healthgeographics.biomedcentral.com/articles/10.1186/1476-072X-10-58

The MAUP is abused for Gerrymandering





Can You Gerrymander Your Party to Power?

By Ella Koeze, Denise Lu and Charlie Smart

Gerrymandering is the intentional distortion of political districts to give one party an advantage, and it has been criticized for disenfranchising many voters and fueling deeper polarization.

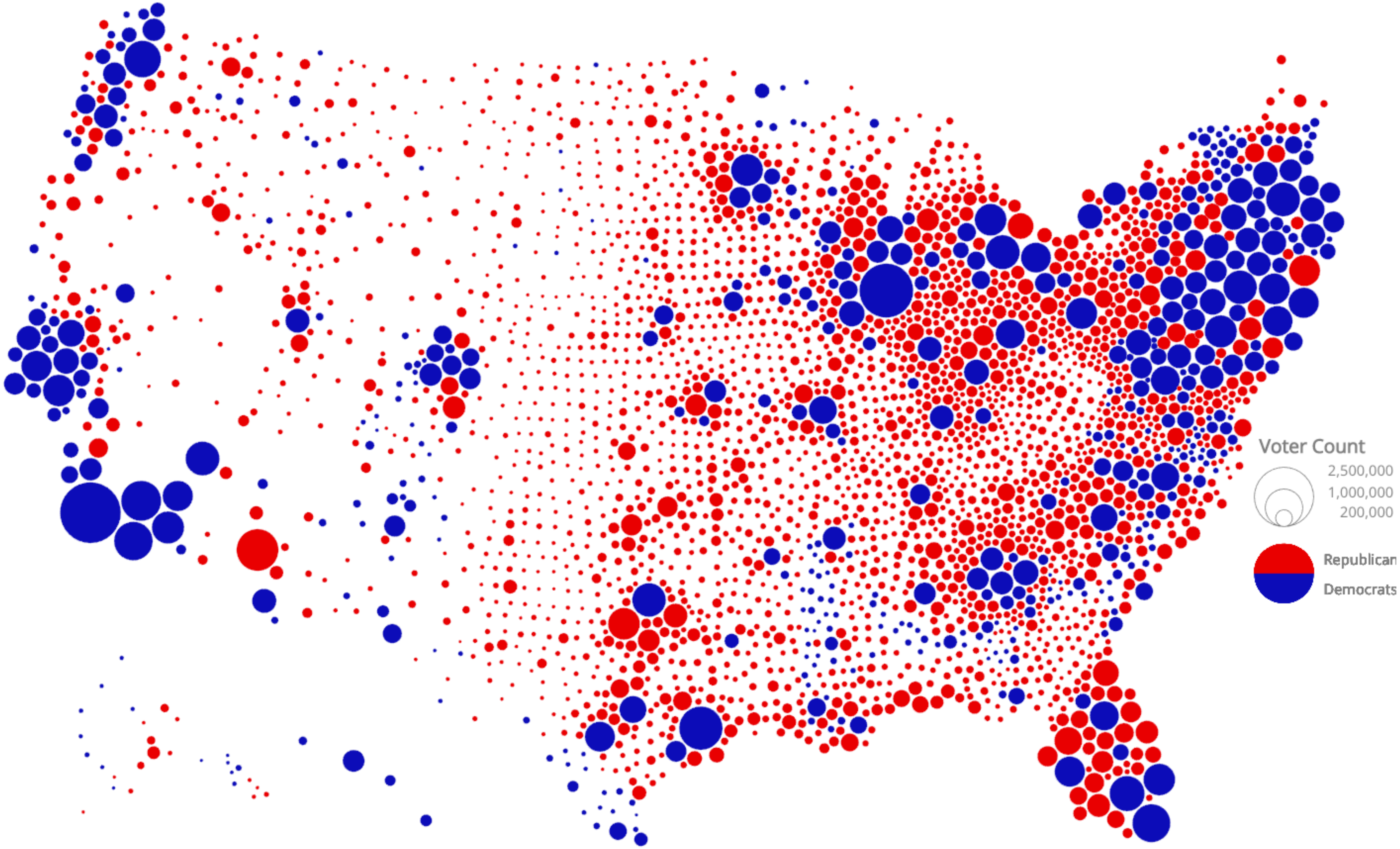
To help you understand it better, we created an imaginary state called **Hexapolis**, where your only mission is to gerrymander your party to power.

https://www.nytimes.com/interactive/2022/01/27/us/politics/congressional-gerrymandering-redistricting-game-2022.html

Related bias: Area is not population

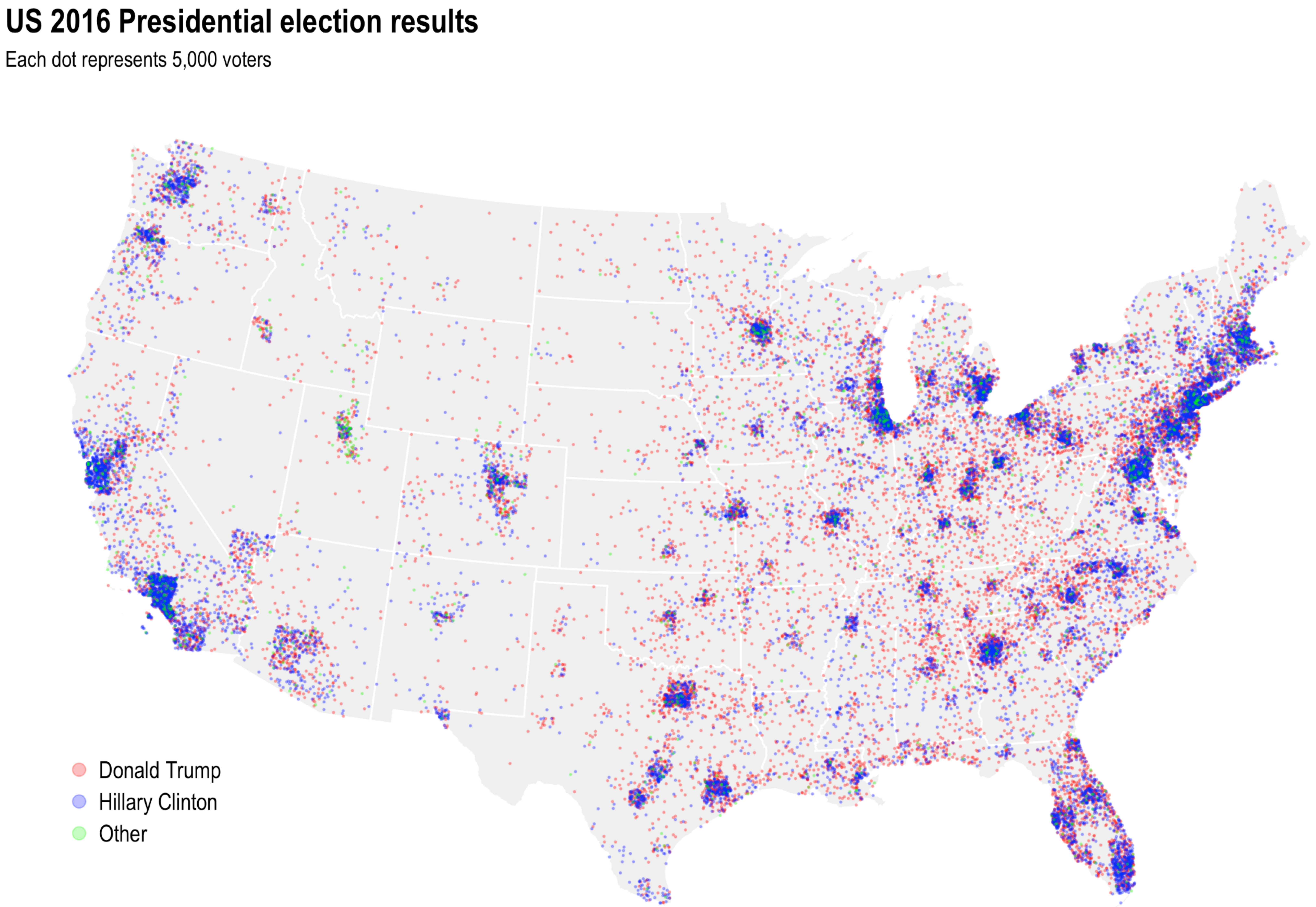
http://try-to-impeach-this.jetpack.ai/

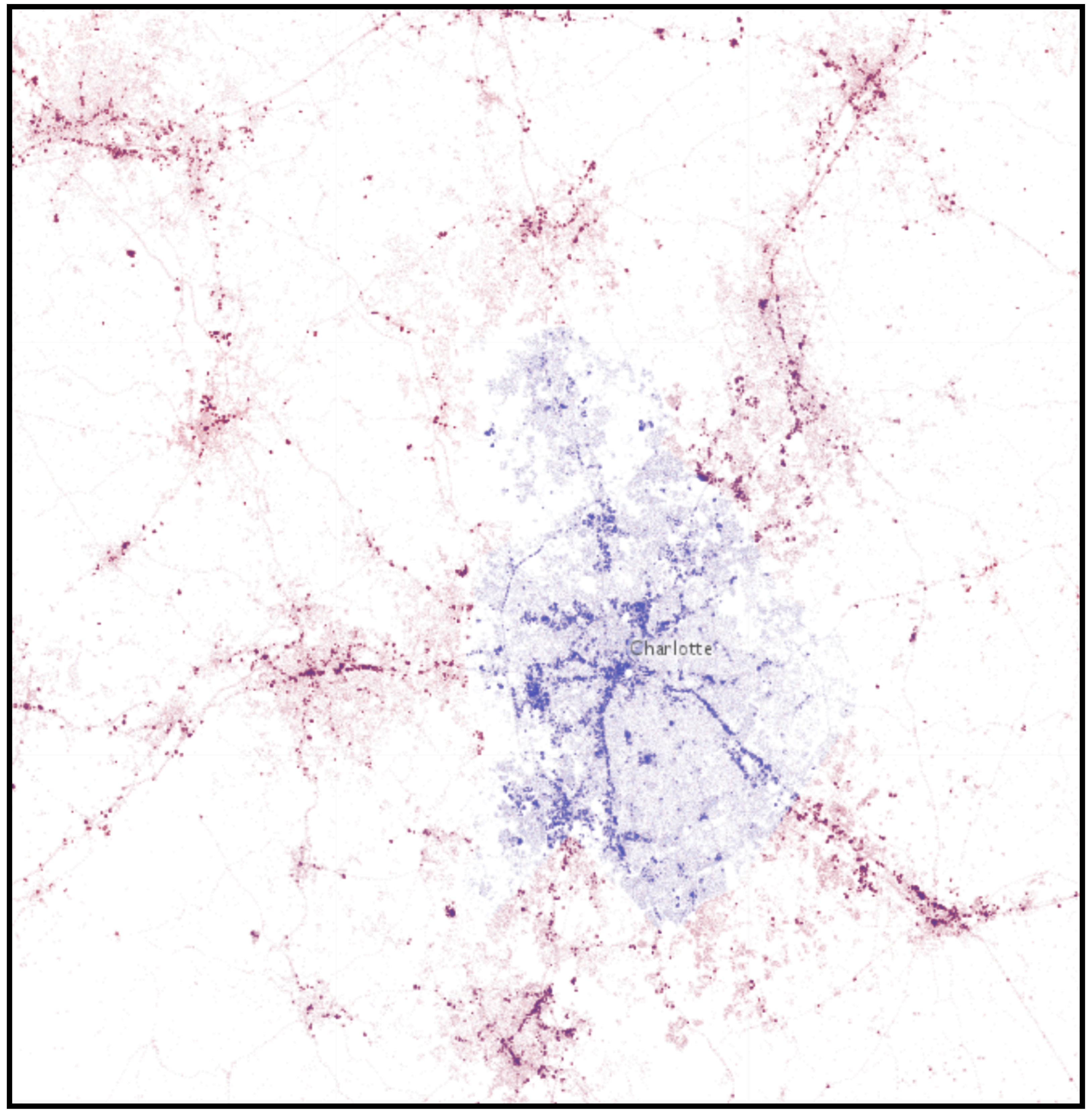
Related bias: Area is not population



http://try-to-impeach-this.jetpack.ai/

A dot density map can reduce the bias





https://carto.maps.arcgis.com/apps/webappviewer/index.html?id=8732c91ba7a14d818cd26b776250d2c3
https://www.maproomblog.com/2018/04/kenneth-fields-dot-density-election-map-redux/

Always mind the MAUP
when exploring
aggregated data

Classification = Transforming continuous into categorical

Student ID	Year	Grade Point Average (GPA)	...
	⋮		
1034262	Senior	3.24	...
1052663	Sophomore	3.51	...
1082246	Freshman	3.62	...
	⋮		

reject
accept
accept

How many categories should there be? If 2: Binarization
How should the values be mapped? What should be the thresholds?

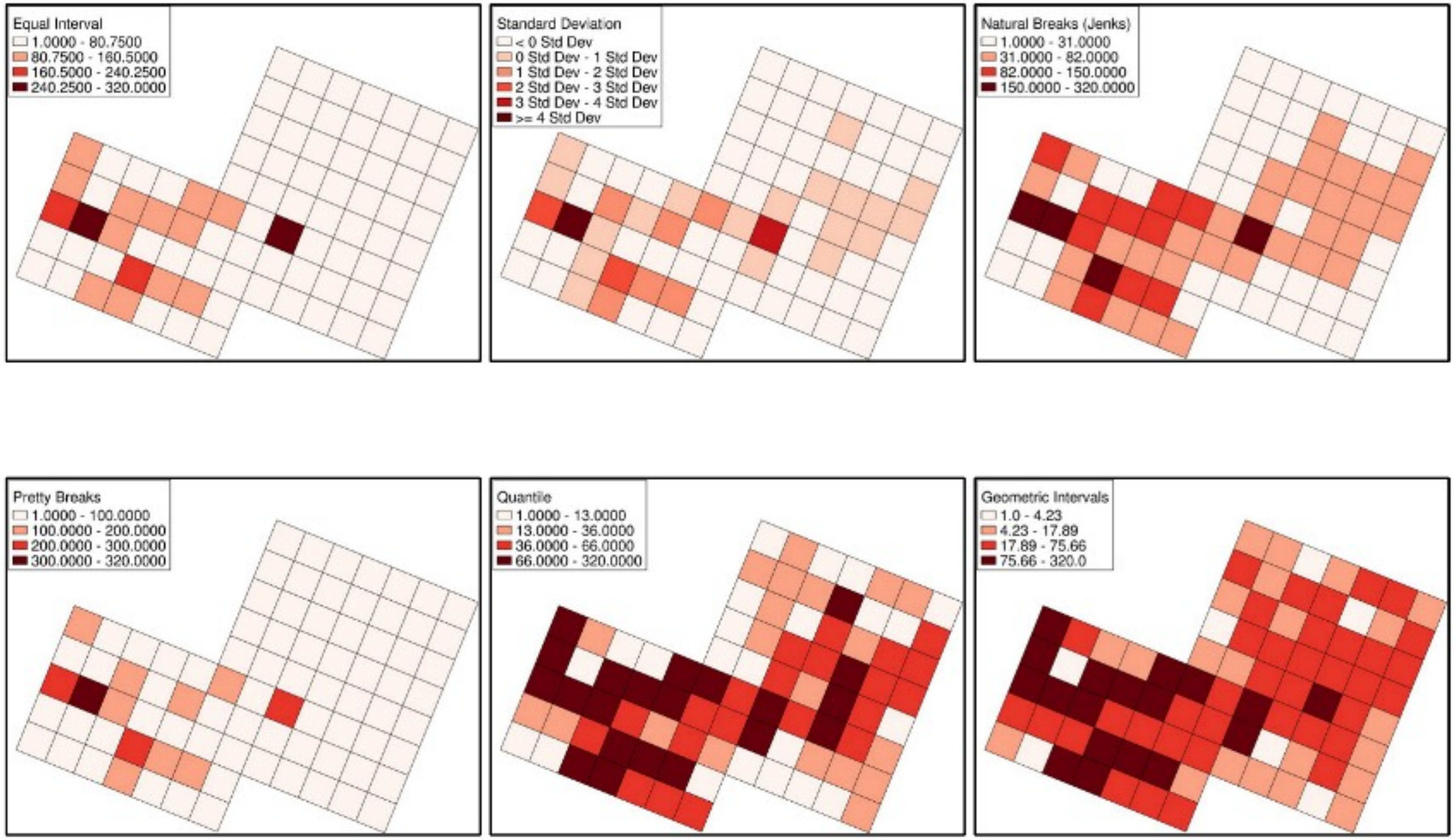
Number of finds in an excavation grid

Group experiment

Each of 3 groups describes their picture in 1 sentence

Data can be classified very differently

Example: Weight of finds in an excavation grid



Same data, different split points

http://184.106.205.13/uselessarchaeology/CMSimple_UA/?Programming___QGIS_Geometric_Class

Classification: Grouping similar phenomena to gain relative simplicity in communication and interpretation. Also known as **binning**.

Formally, you must select thresholds to classify your data

The classification problem is to define class boundaries such that

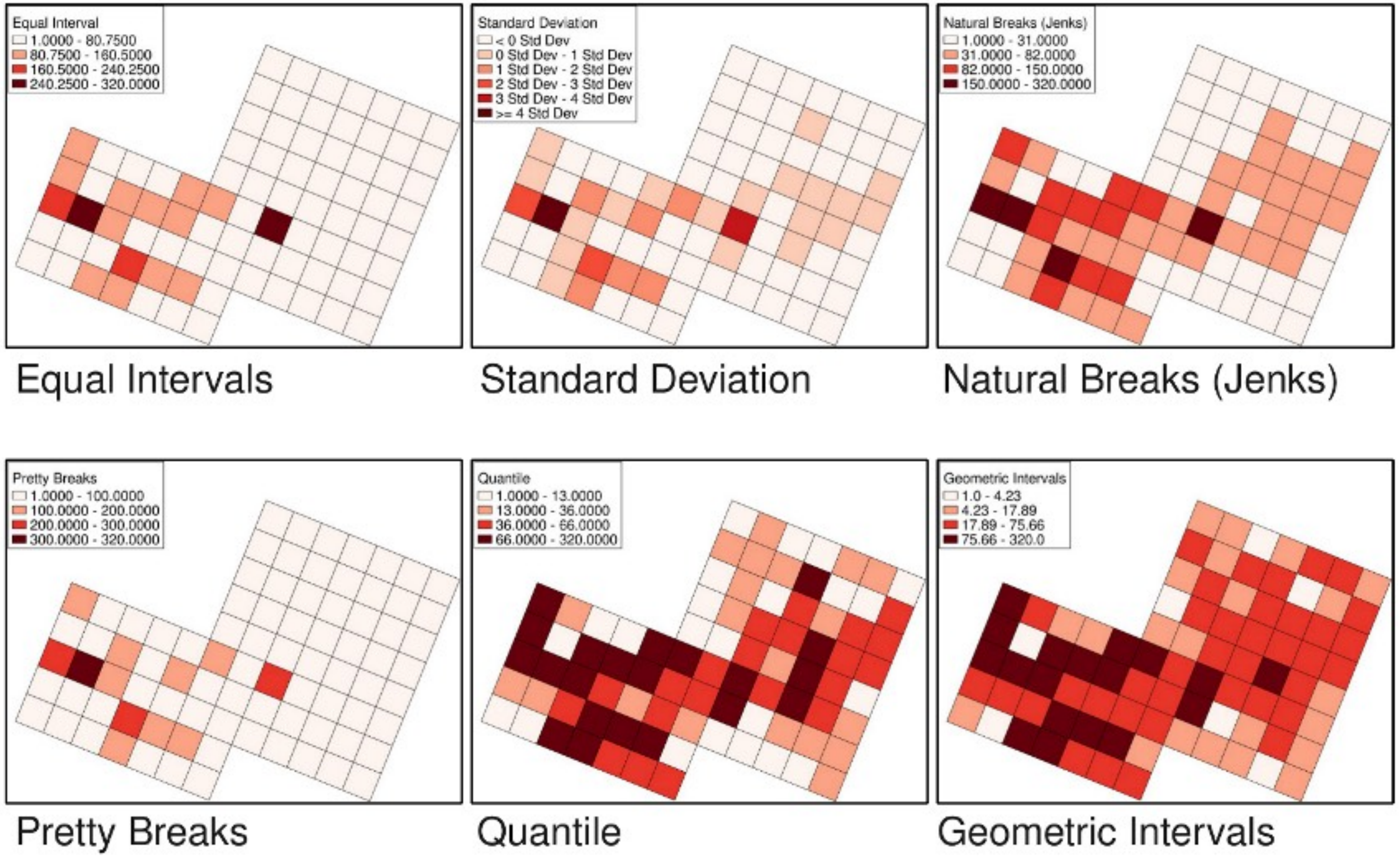
$$c_j < y_i \leq c_{j+1} \ \forall y_i \in C_j$$

where y_i is the value of the attribute for spatial location i,
j is a class index,
c_j represents the lower bound of interval j.

The choice of class boundaries defines the classification scheme

mapclassify:

'Head-Tail Breaks',
'Fisher-Jenks',
'Maximum Breaks',
'Equal Interval',
'MaxP',
'Quantiles',
'Jenks Caspall',
'Mean-Standard Deviation'

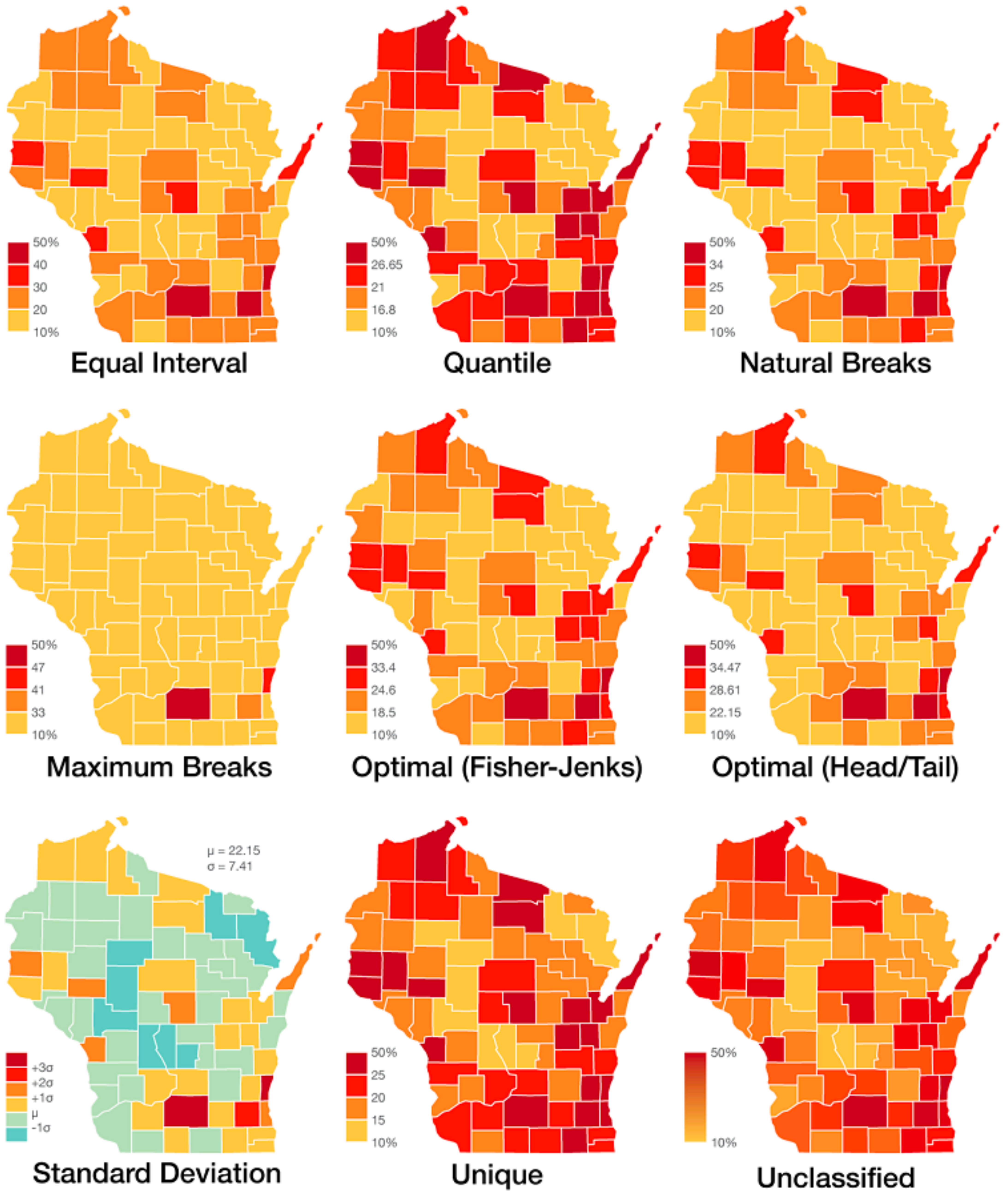


https://pypi.org/project/mapclassify/

Data can be classified very differently

Same data,
Same spatial units,
Same colors,
Different classification

Percentage of residents over 25 with a Bachelor's degree



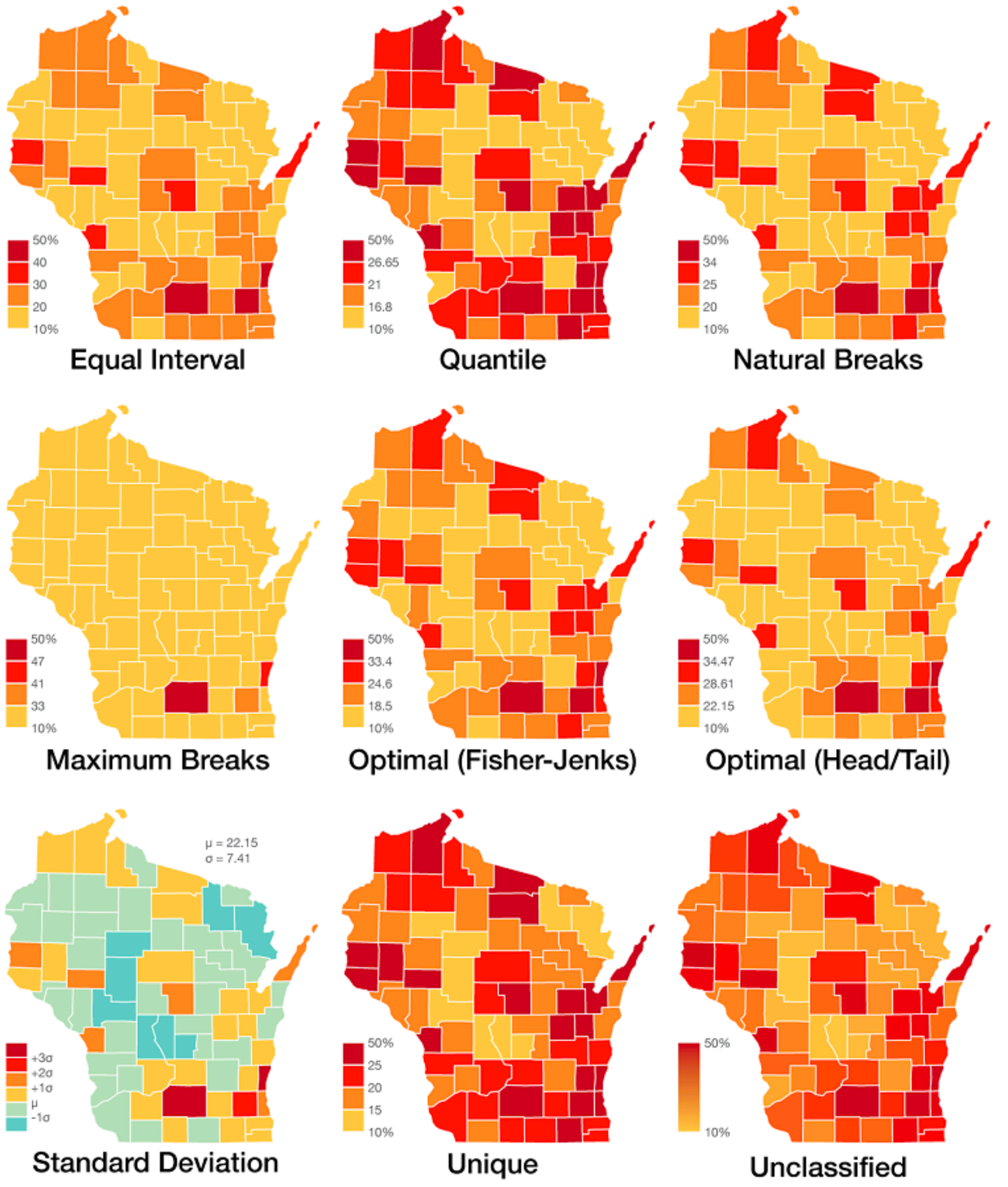
https://gistbok.ucgis.org/bok-topics/statistical-mapping-enumeration-normalization-classification

Data can be classified very differently

A data set does not have a "perfect" choropleth map

As with MAUP, you can create *arbitrary* interpretations of the same data!

Percentage of residents over 25 with a Bachelor's degree



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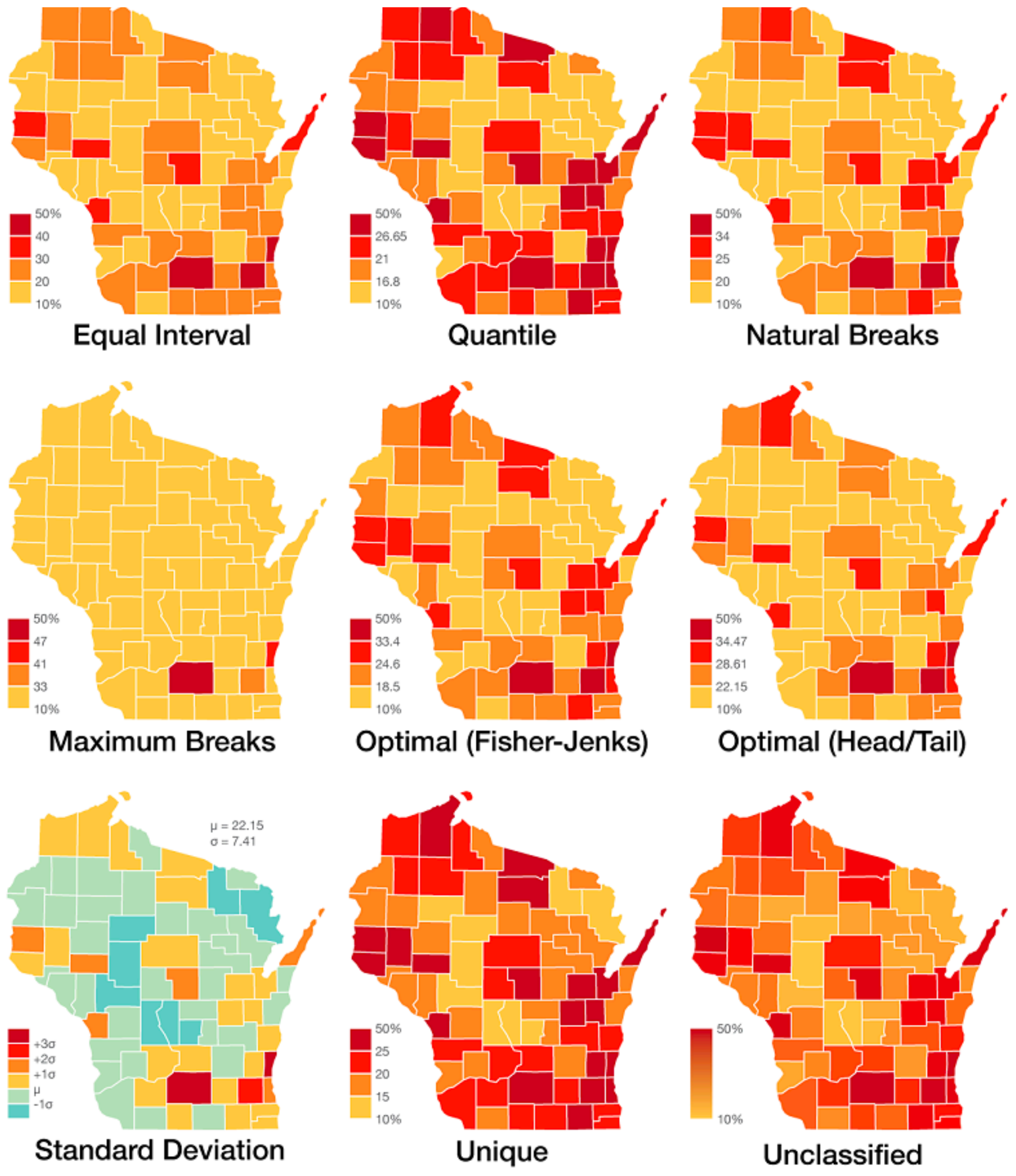
The choice of class boundaries defines the classification scheme

mapclassify:

```
'EqualInterval',
'FisherJenks',
…
'HeadTailBreaks',
'JenksCaspall',
…
'MaximumBreaks',
'NaturalBreaks',
'Percentiles',
…
'Quantiles',
'StdMean',
'UserDefined',
```

Percentage of residents over 25 with a Bachelor's degree

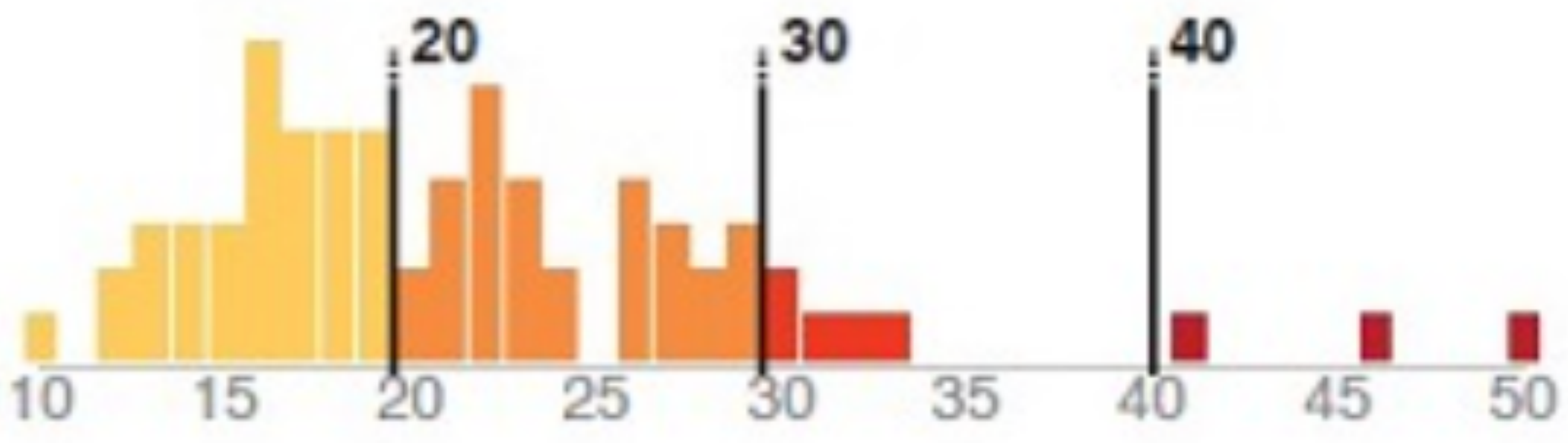


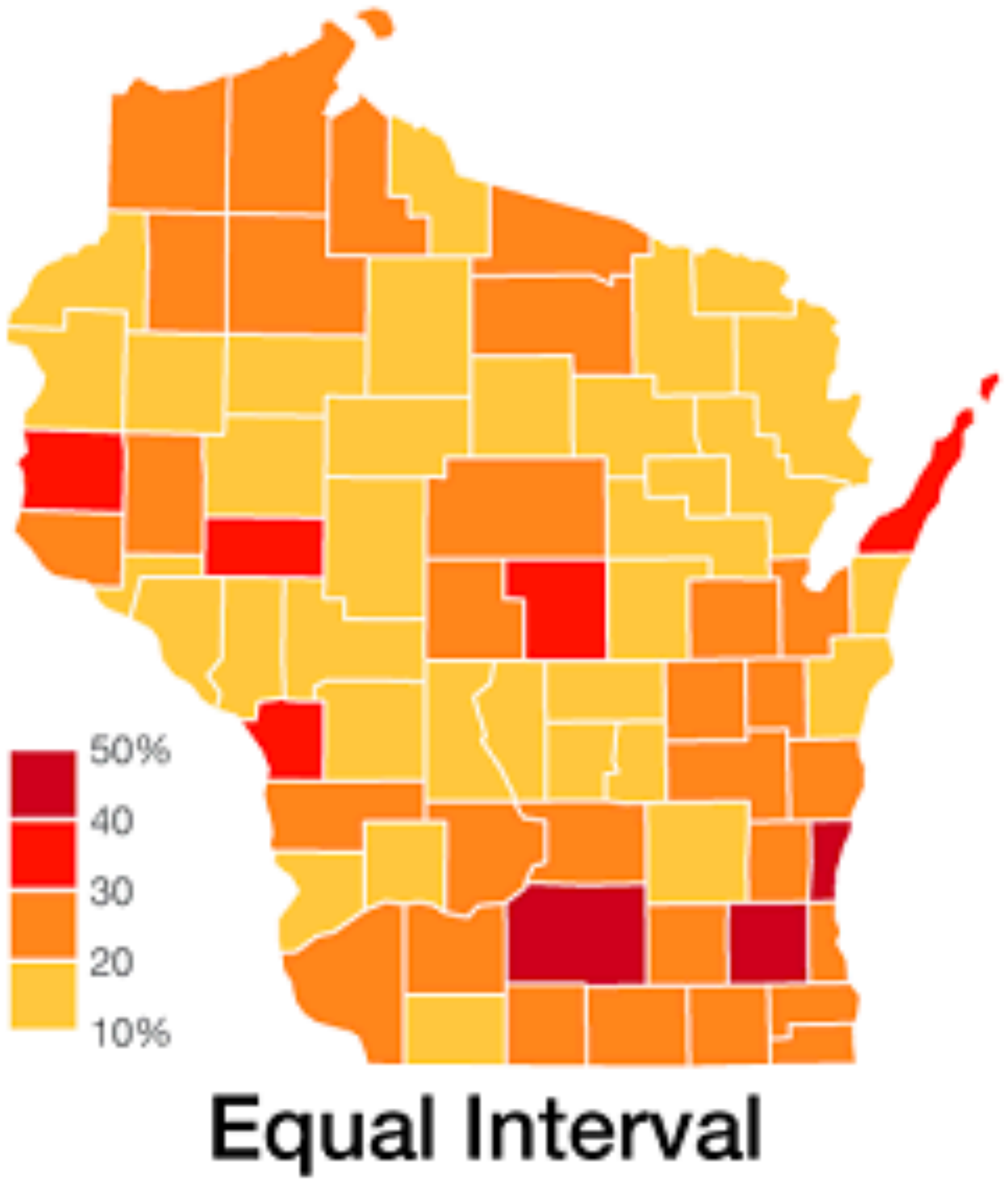
https://pypi.org/project/mapclassify/

https://gistbok.ucgis.org/bok-topics/statistical-mapping-enumeration-normalization-classification

Equal Interval

For uniformly distributed data with familiar data ranges.

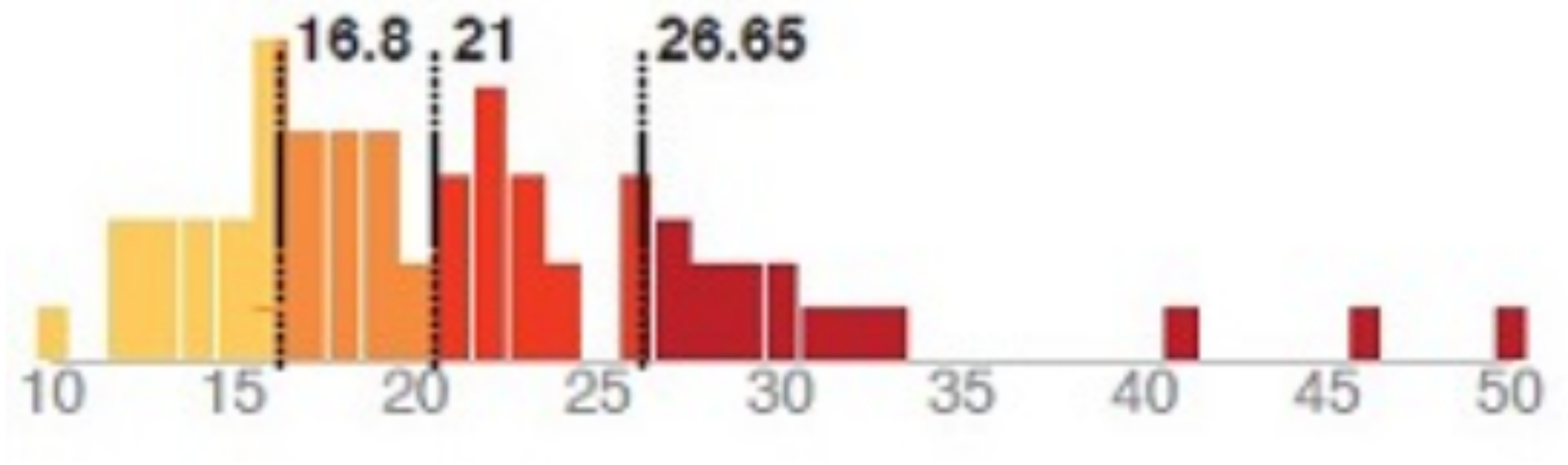


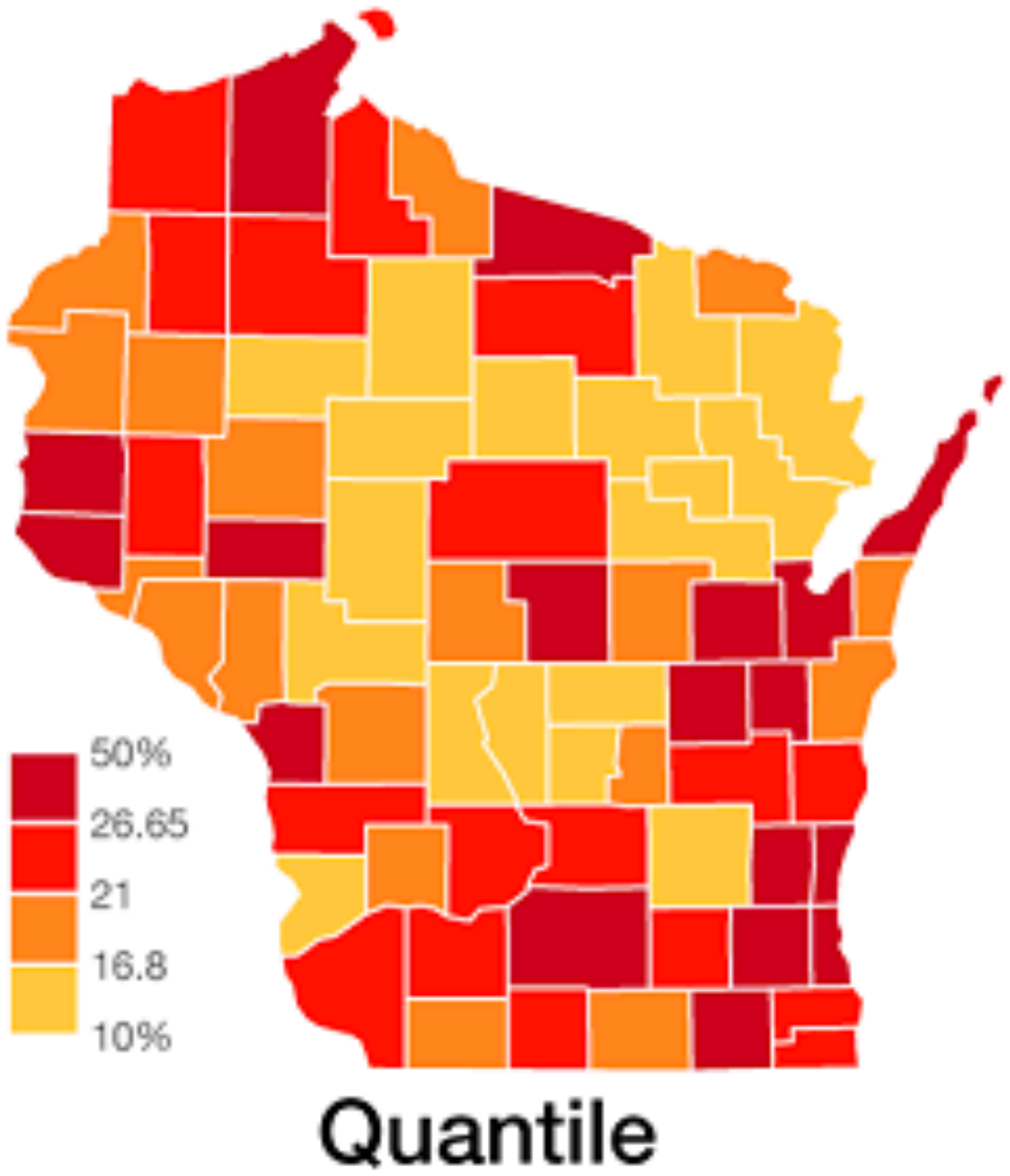


https://gistbok.ucgis.org/bok-topics/statistical-mapping-enumeration-normalization-classification

Quantiles (Equal Count)

For evenly distributed data and ordinal data

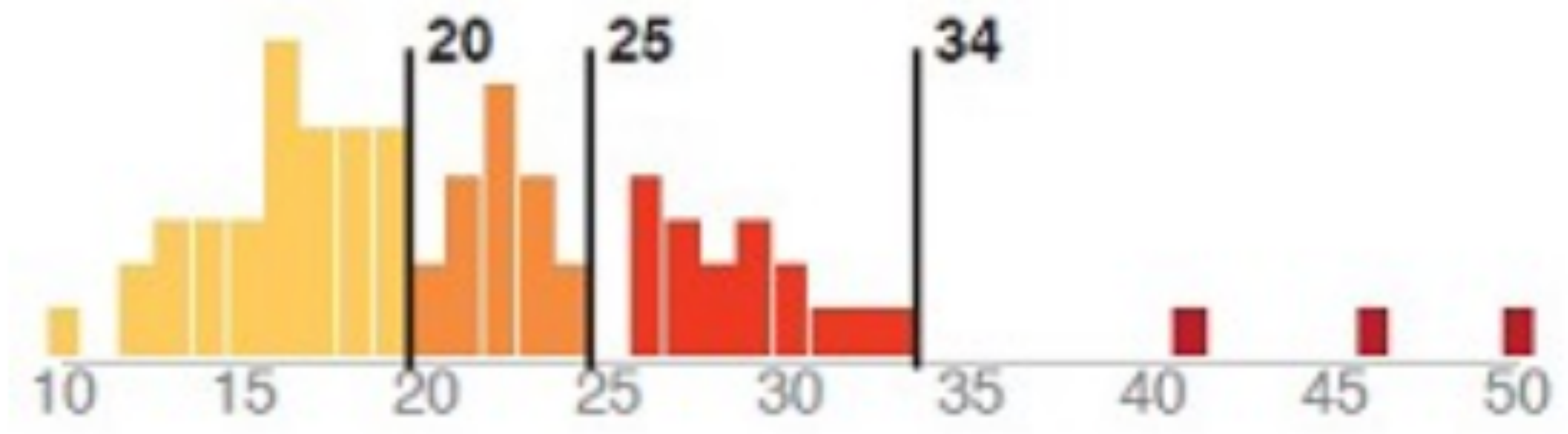


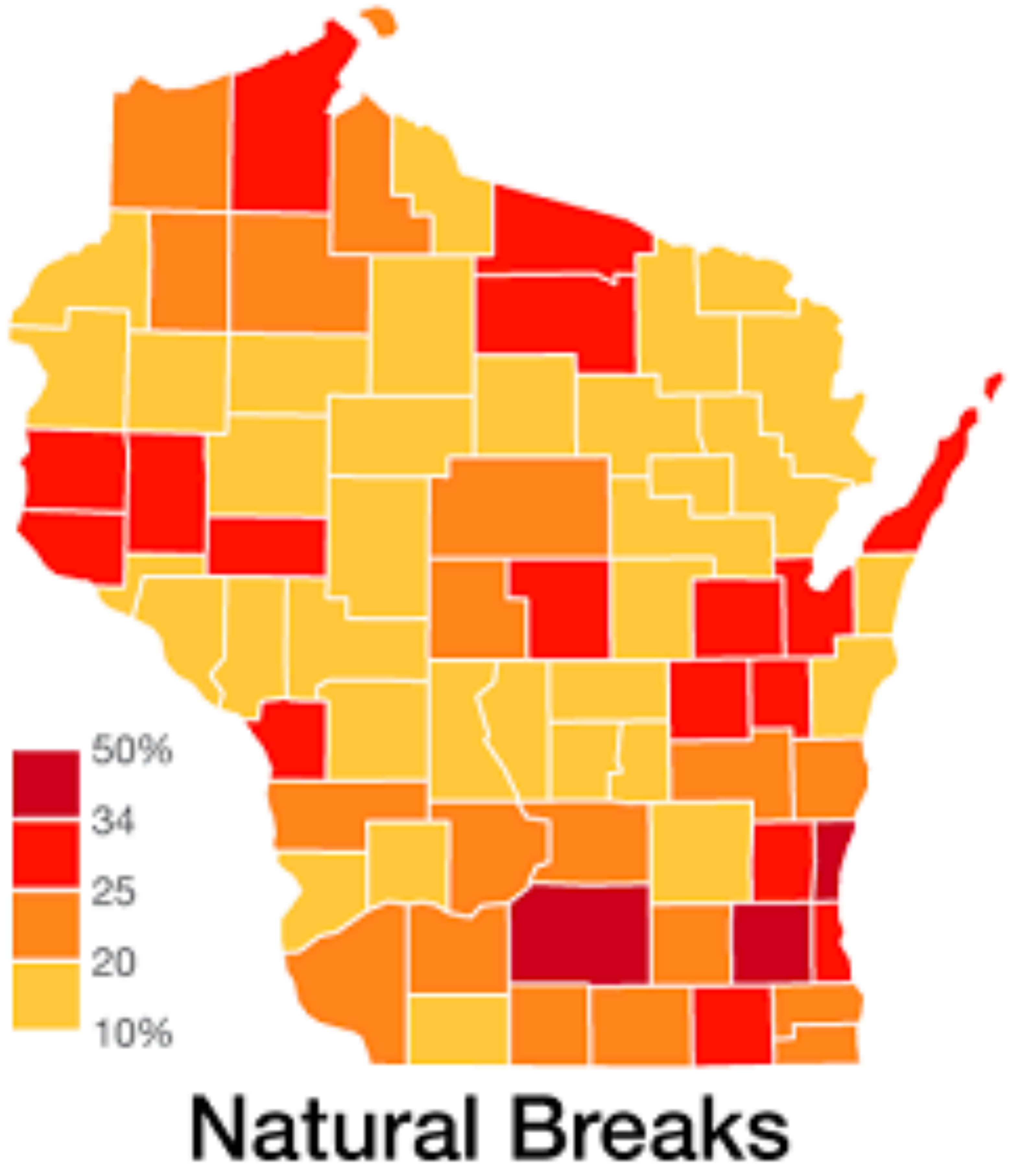


https://gistbok.ucgis.org/bok-topics/statistical-mapping-enumeration-normalization-classification

Natural Breaks

For clustered data

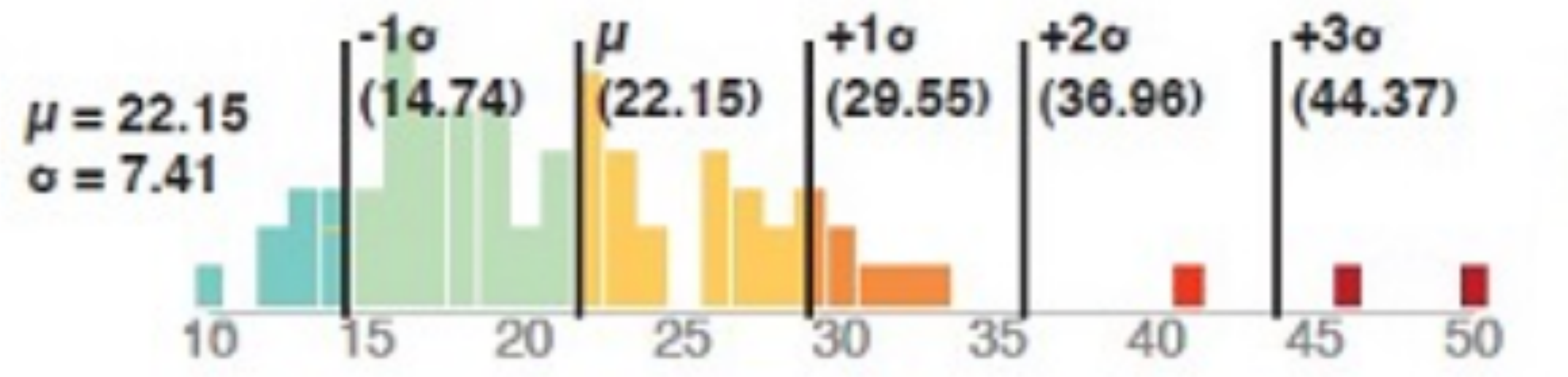


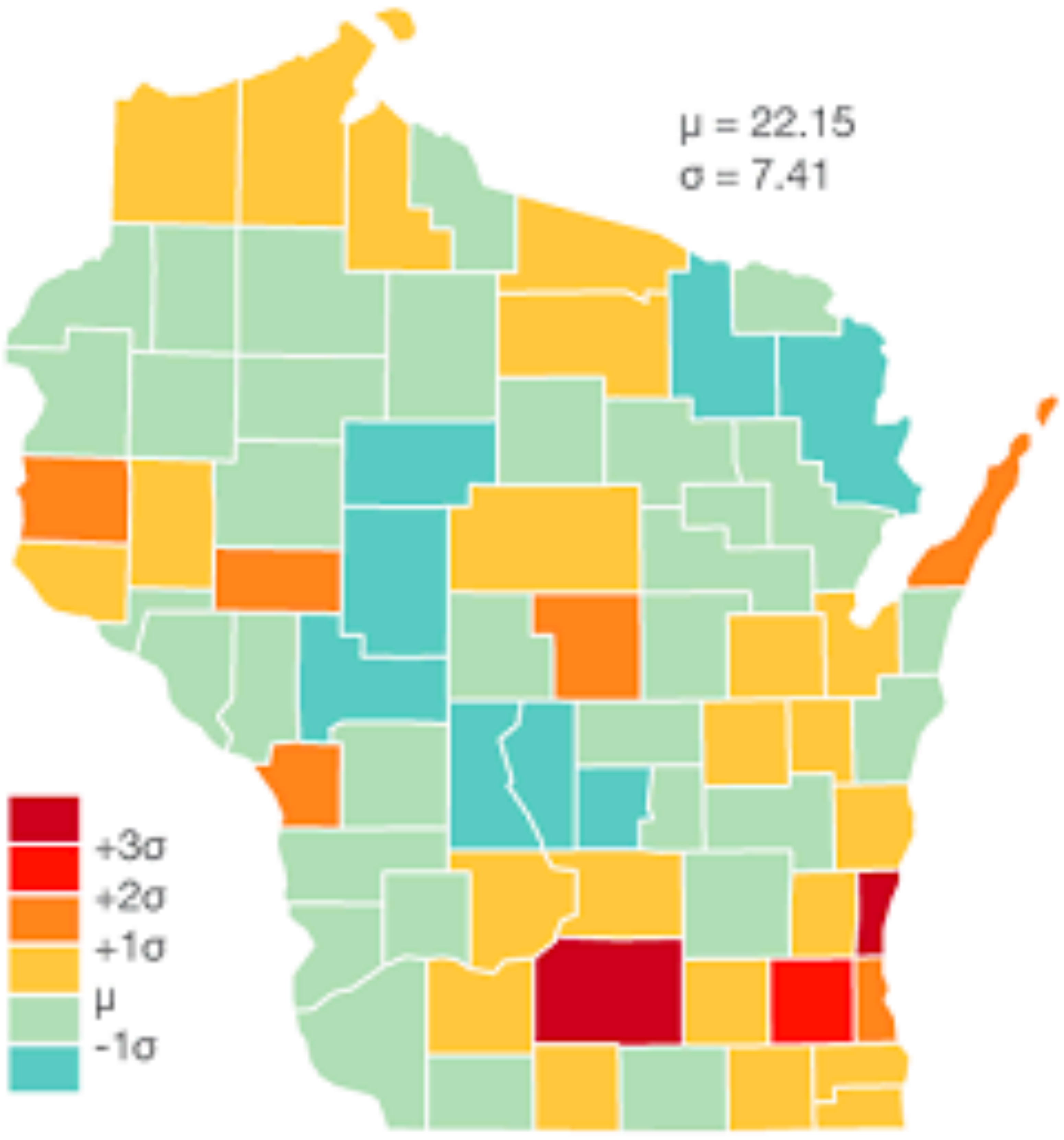


https://gistbok.ucgis.org/bok-topics/statistical-mapping-enumeration-normalization-classification

Mean-Standard Deviation

For normally distributed data



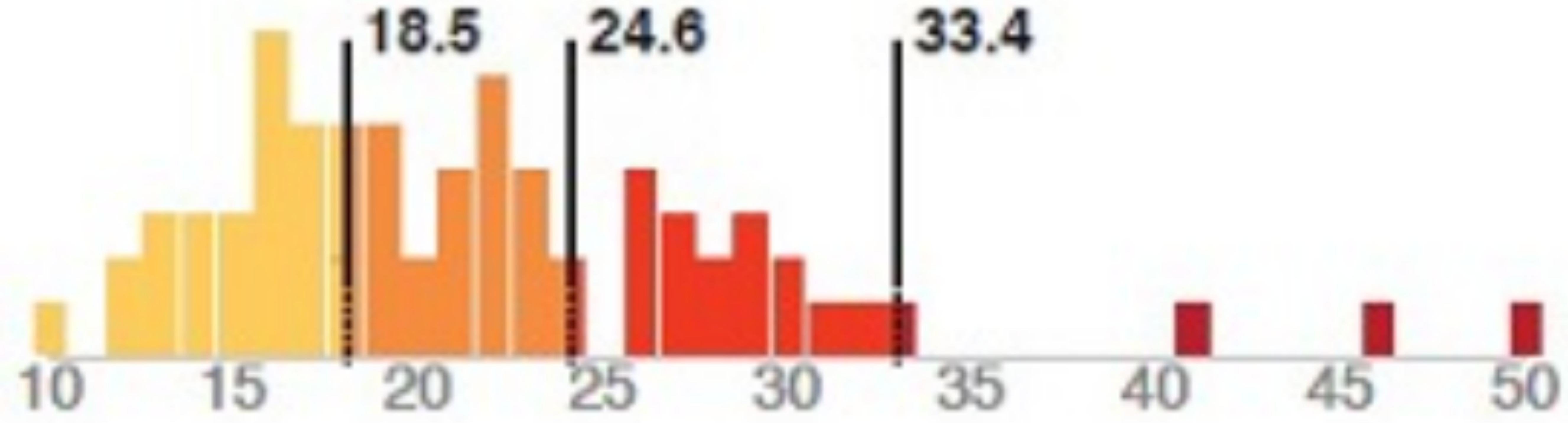


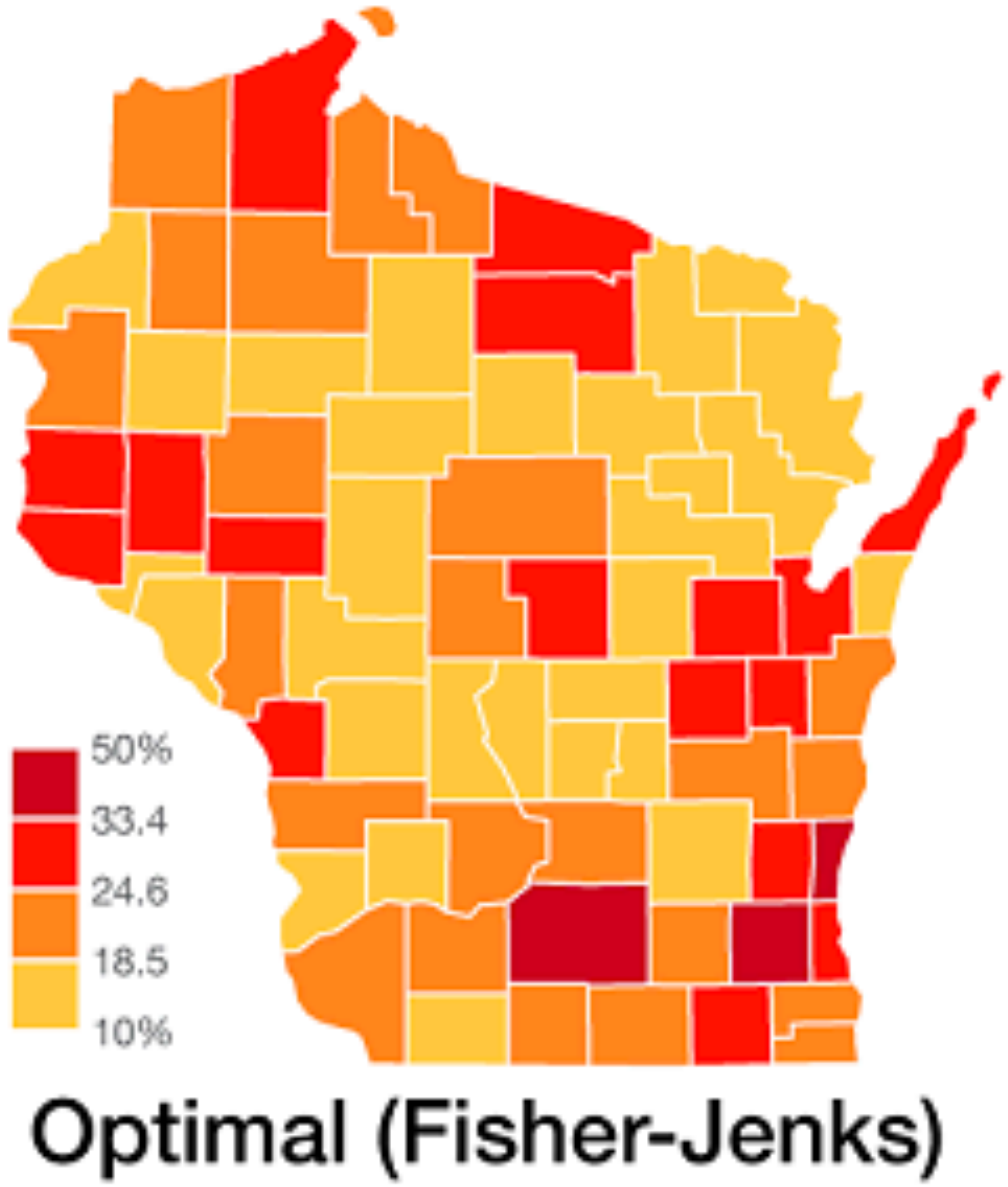
Standard Deviation

https://gistbok.ucgis.org/bok-topics/statistical-mapping-enumeration-normalization-classification

Jenks-Caspall & Fisher-Jenks

For clustered and skewed data

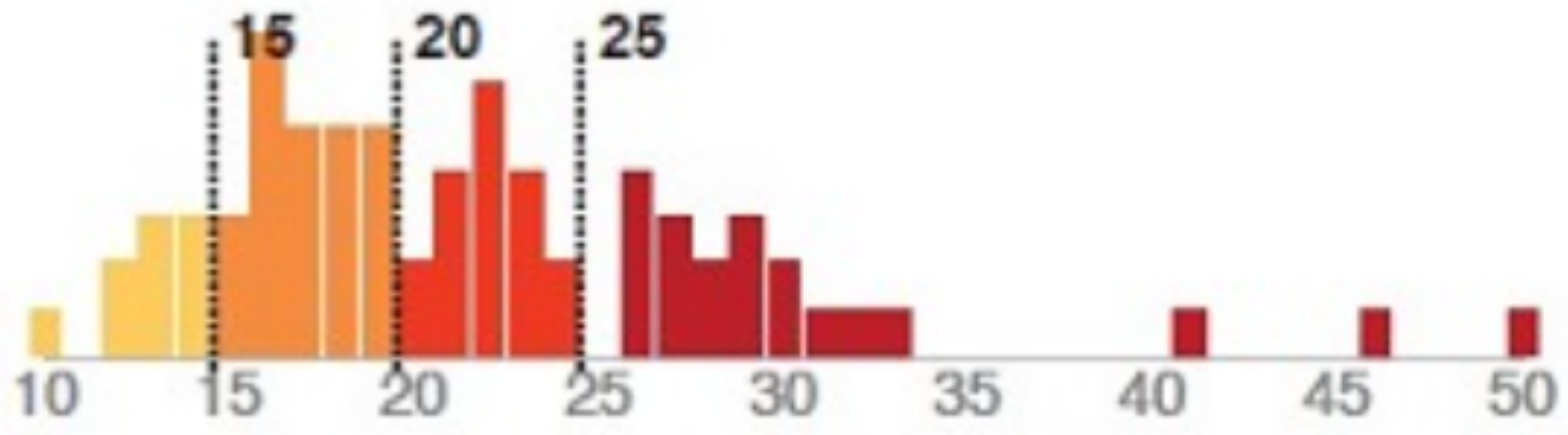


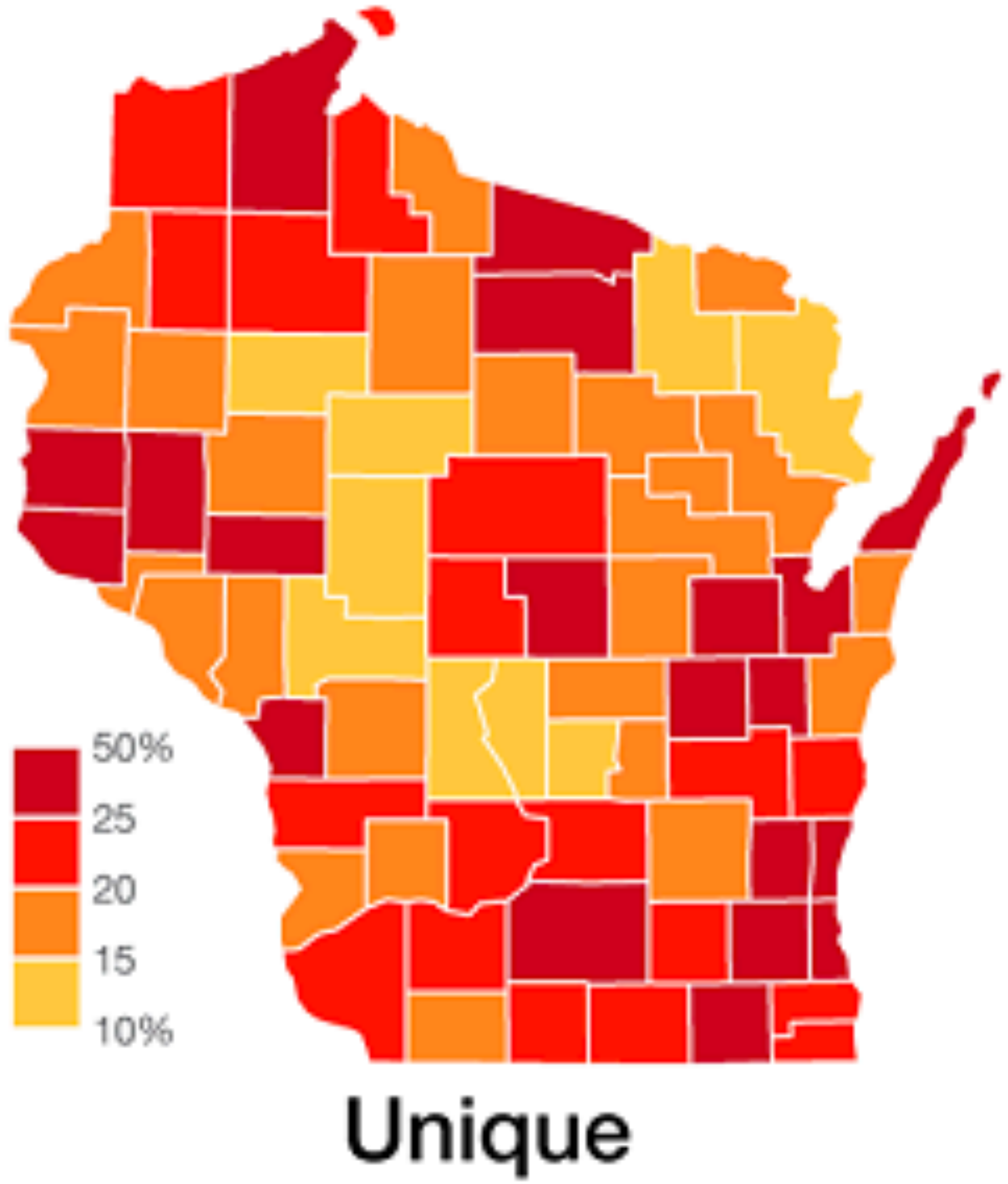


https://gistbok.ucgis.org/bok-topics/statistical-mapping-enumeration-normalization-classification

Unique/Manual

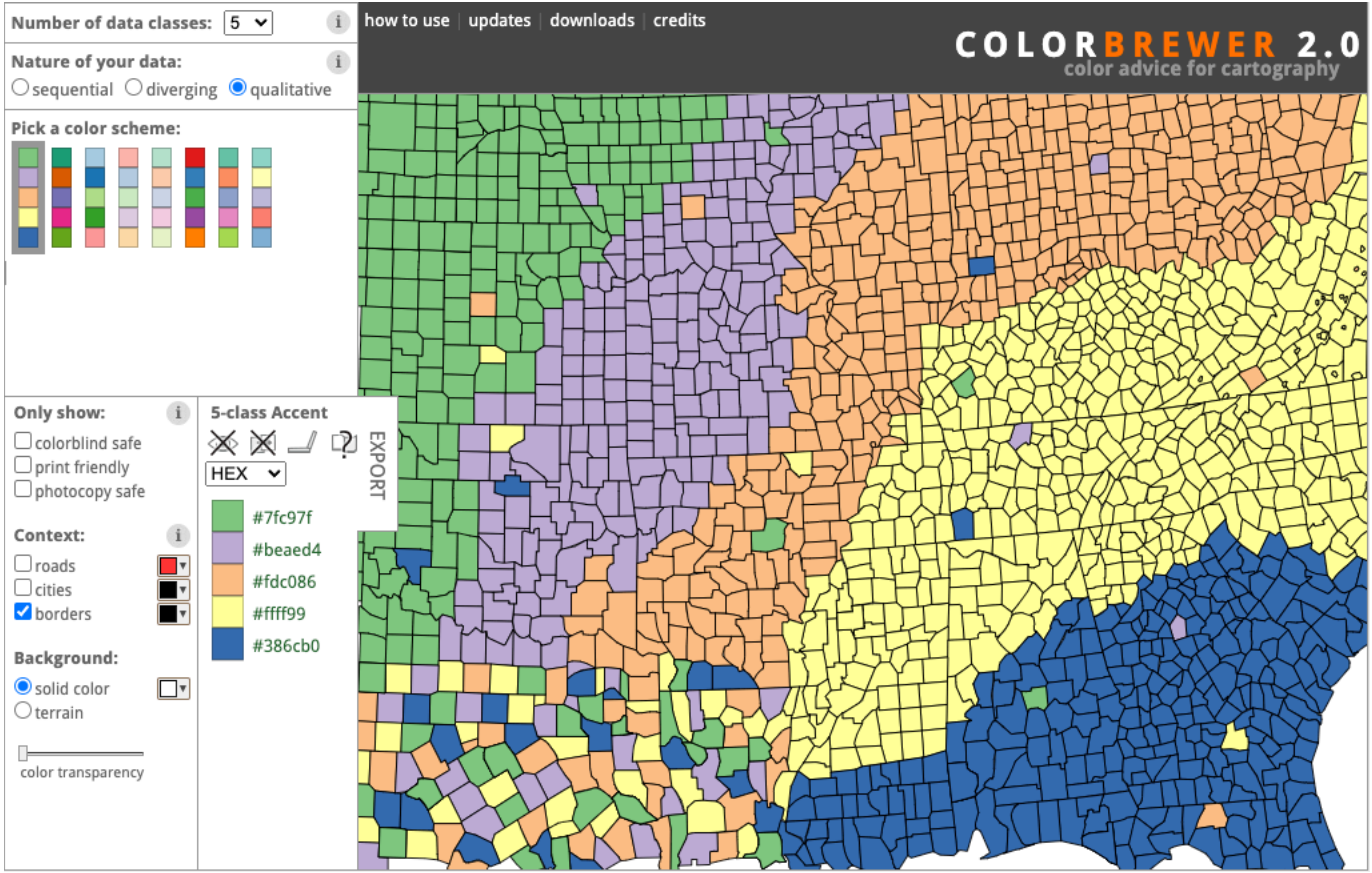
For data where key numbers are important





https://gistbok.ucgis.org/bok-topics/statistical-mapping-enumeration-normalization-classification

For choice of colors use tools like color brewer

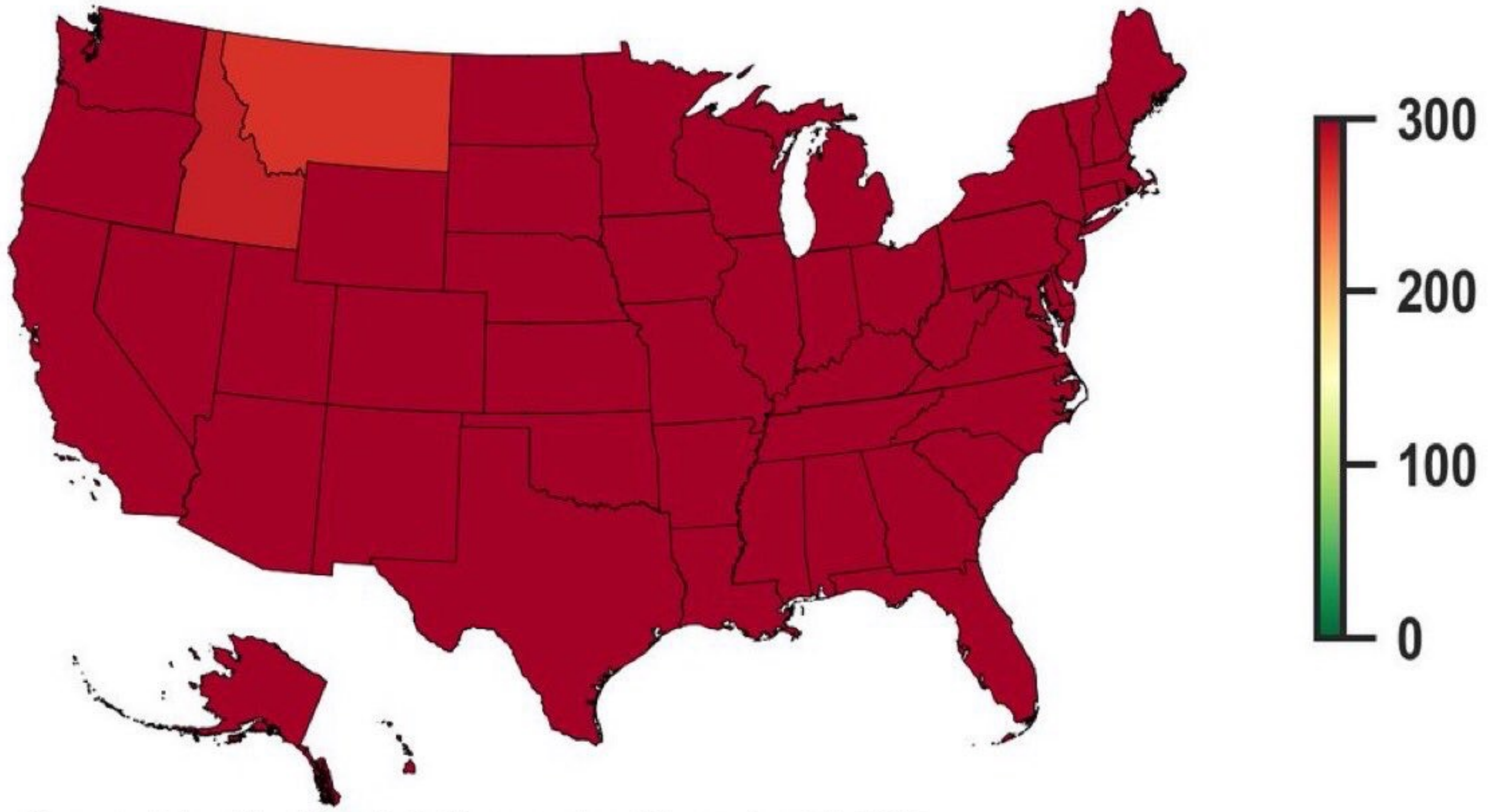


https://colorbrewer2.org/

Example: Wrong use of colors and scales

How the year started

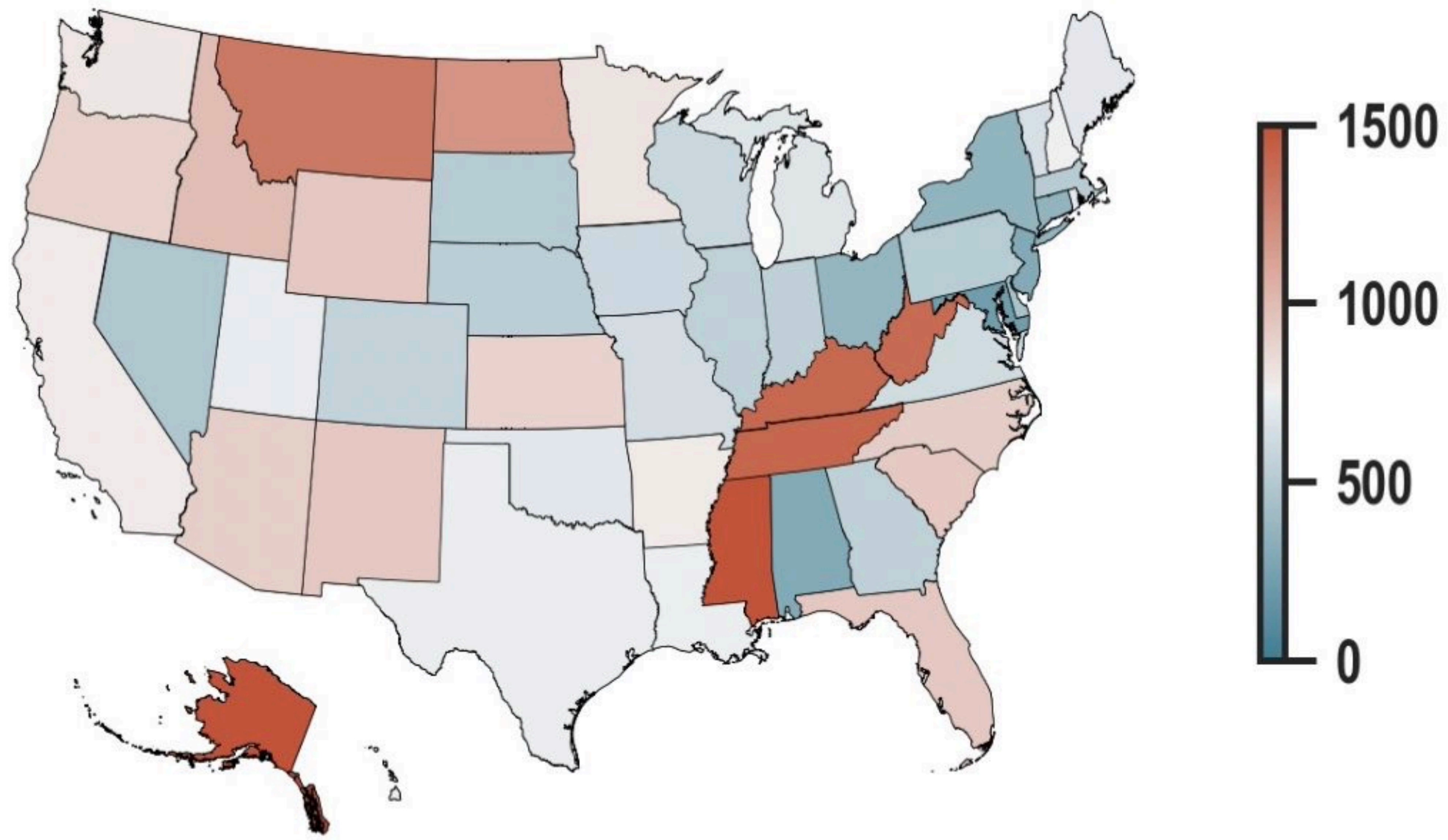
New COVID-19 cases per million population



Source: Johns Hopkins, J. P. Morgan. As of December 31, 2021

How it is going

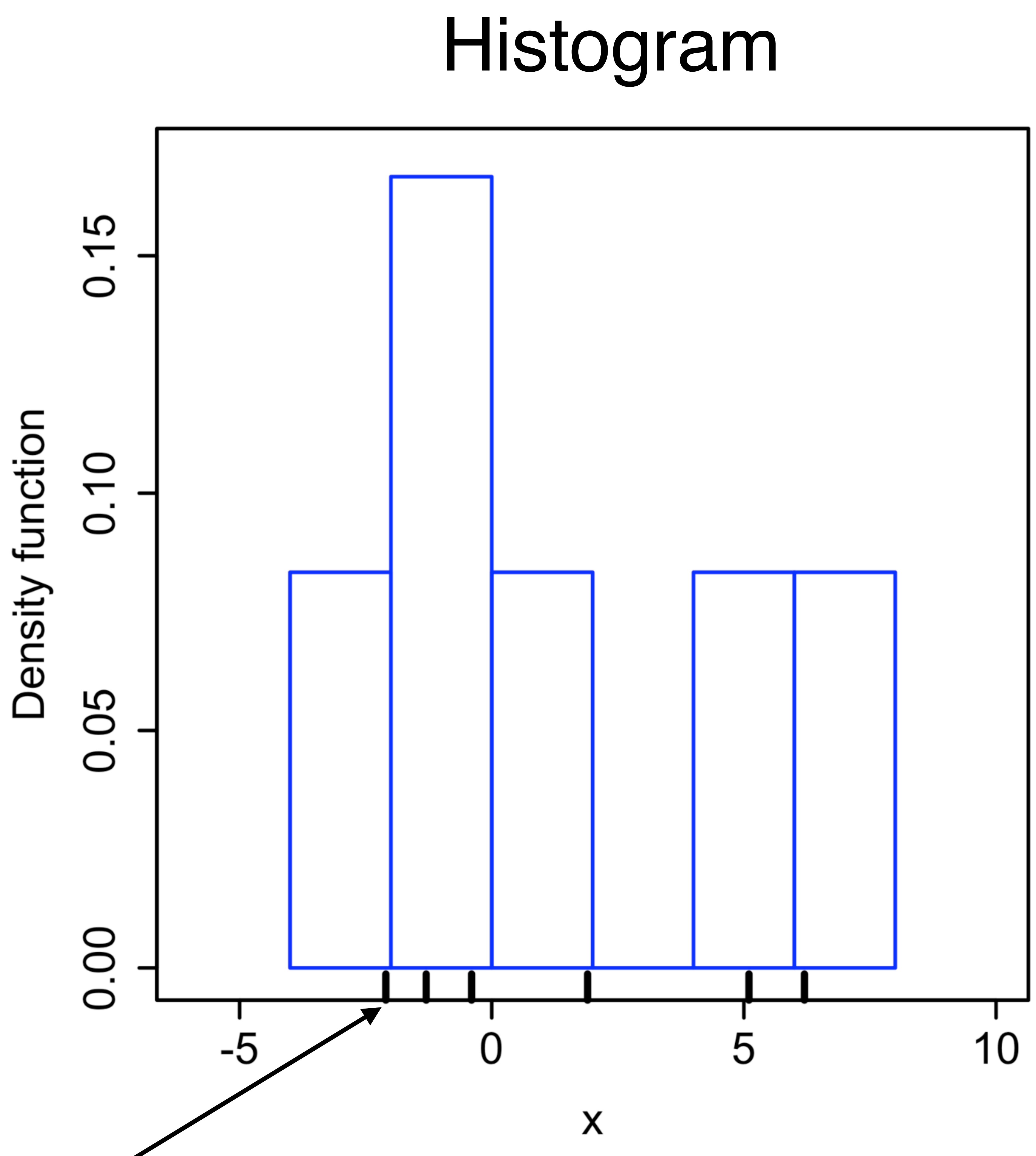
New COVID-19 cases per million population



Source: Johns Hopkins, J. P. Morgan. As of February 08, 2022

https://twitter.com/carlquintanilla/status/1491824980648247296?s=20

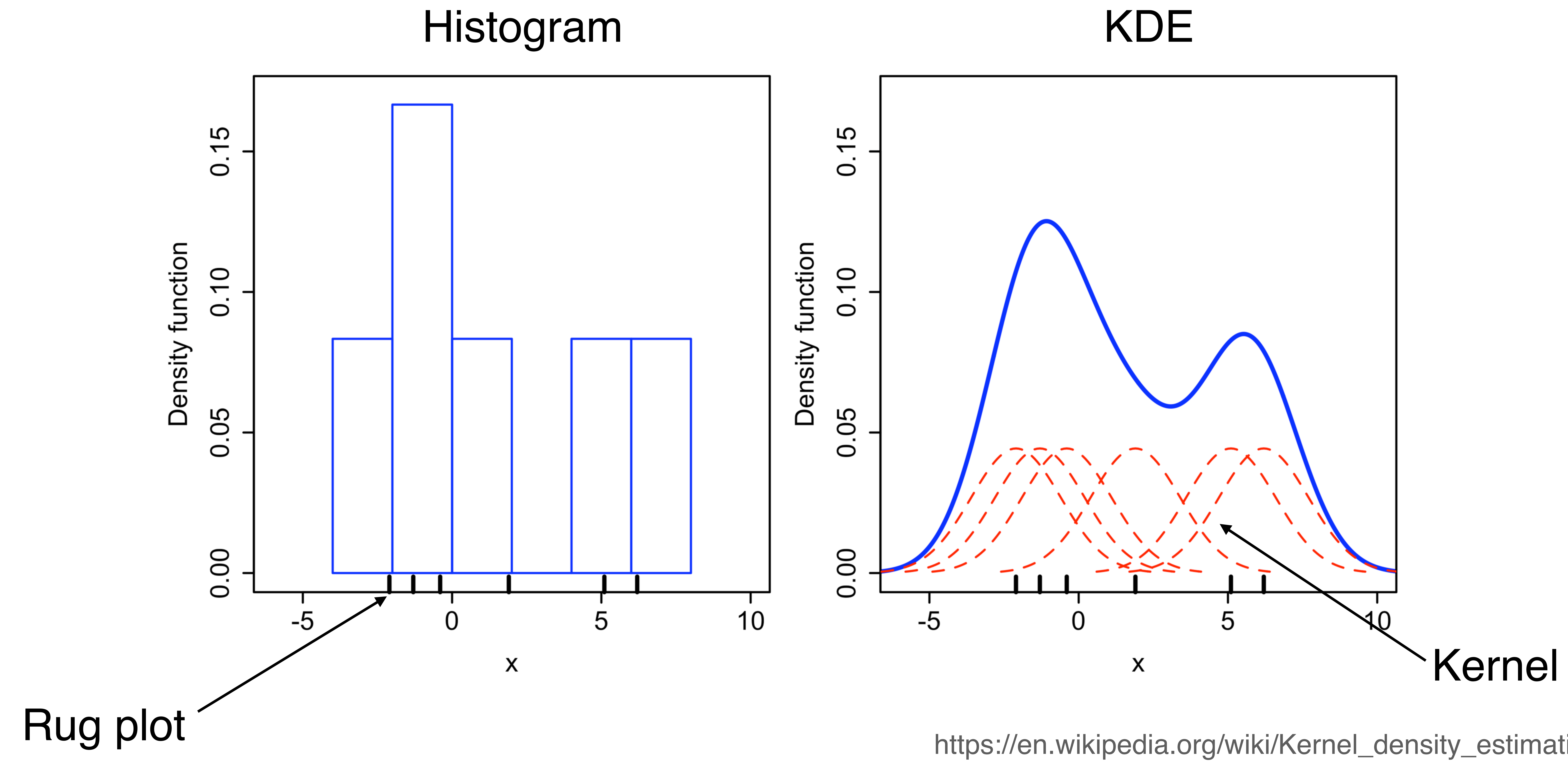
Kernel density estimation (KDE)



Rug plot

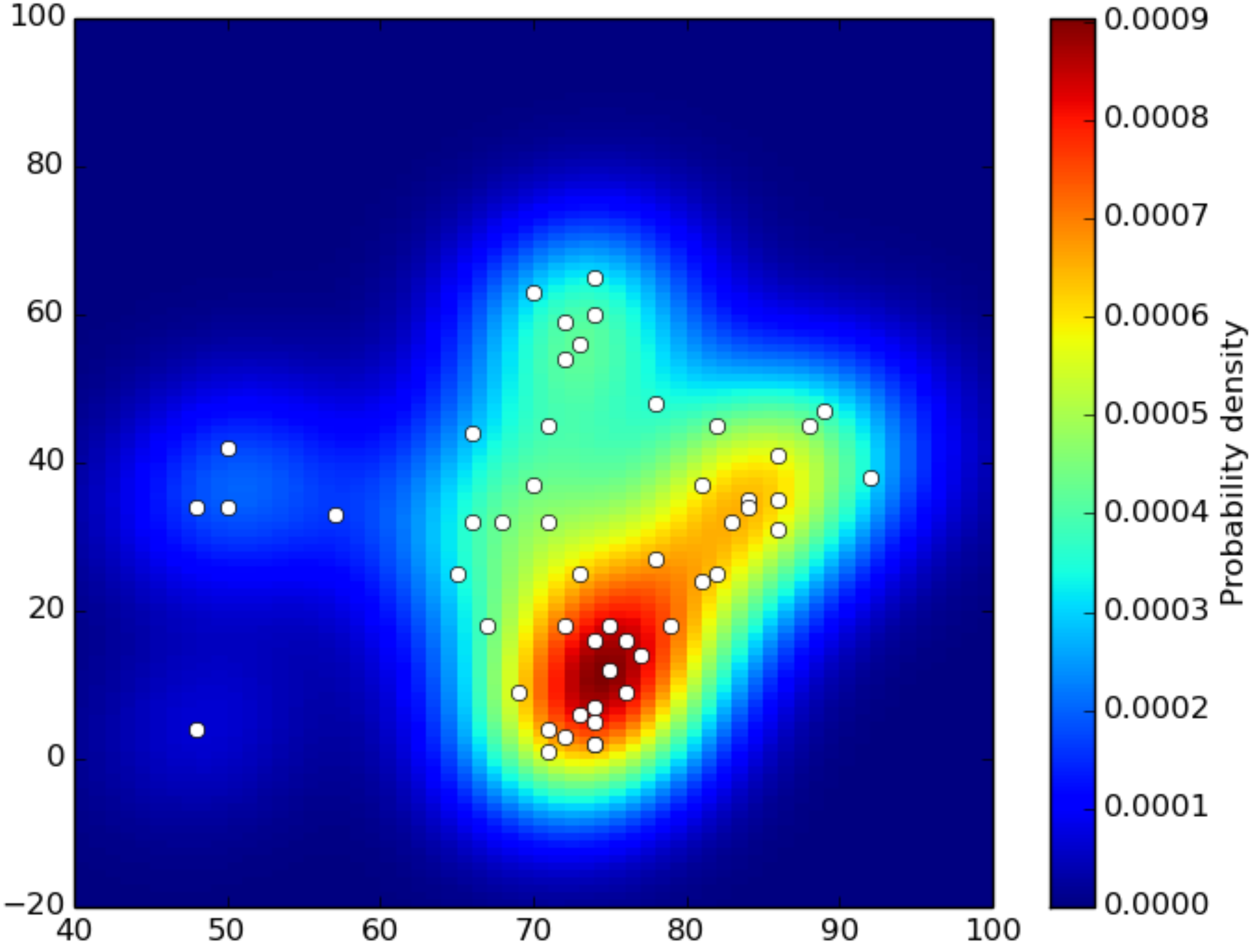
https://en.wikipedia.org/wiki/Kernel_density_estimation

Kernel density estimation creates a smooth pdf to data points



https://en.wikipedia.org/wiki/Kernel_density_estimation

Heatmaps use KDE for geospatial data exploration



Jupyter

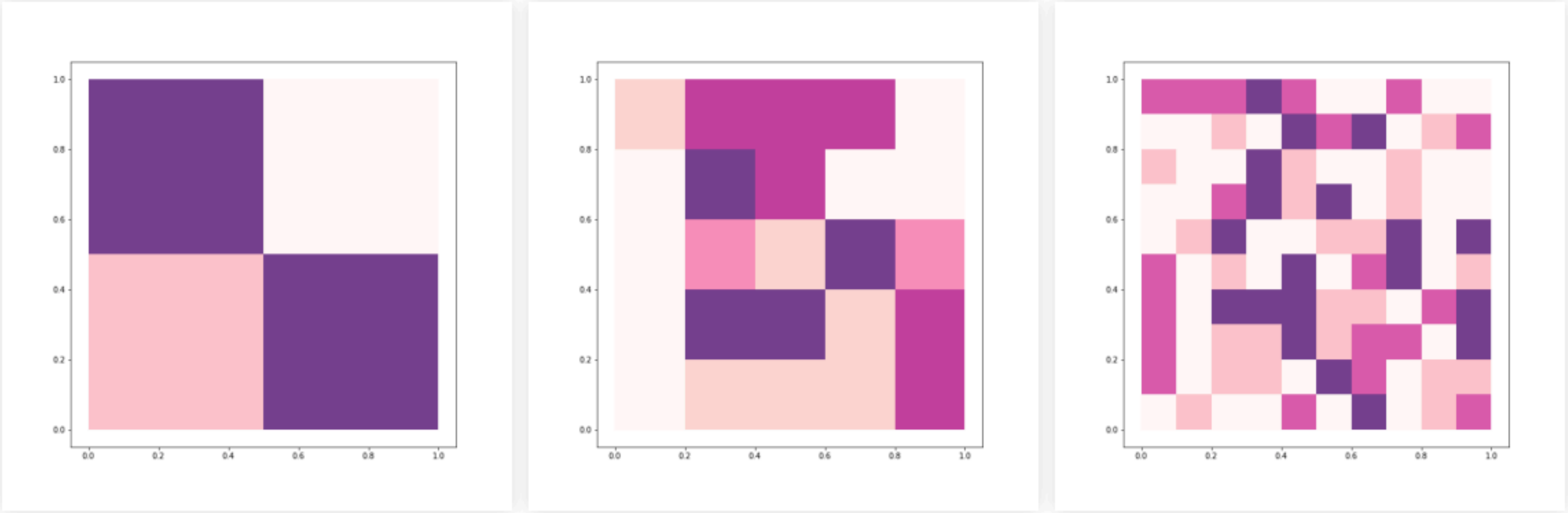
Sources and further materials for today's class

https://geographicdata.science/book/notebooks/05_choropleth.html

https://darribas.org/gds_course/content/bD/concepts_D.html

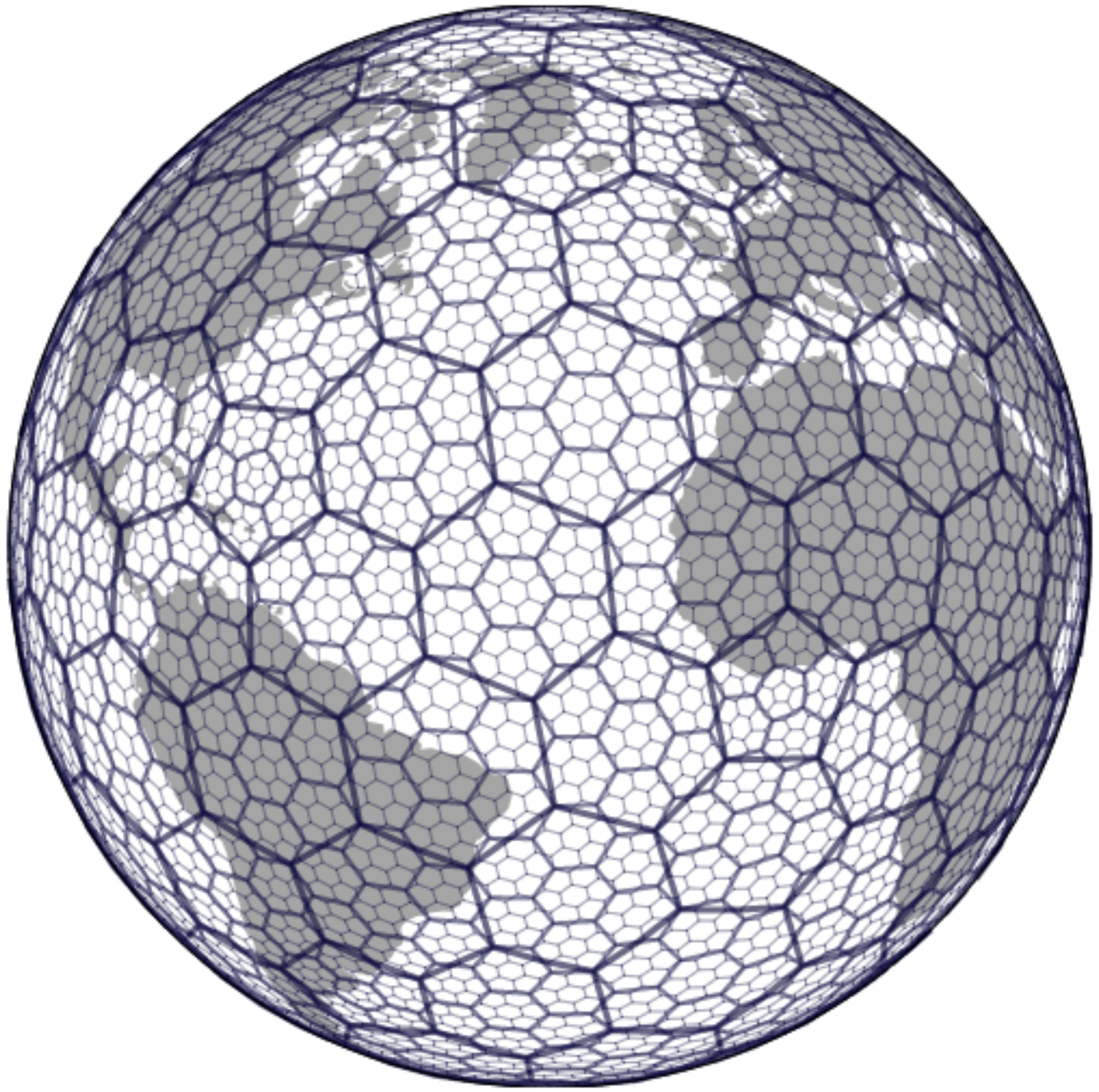
Take home messages for today

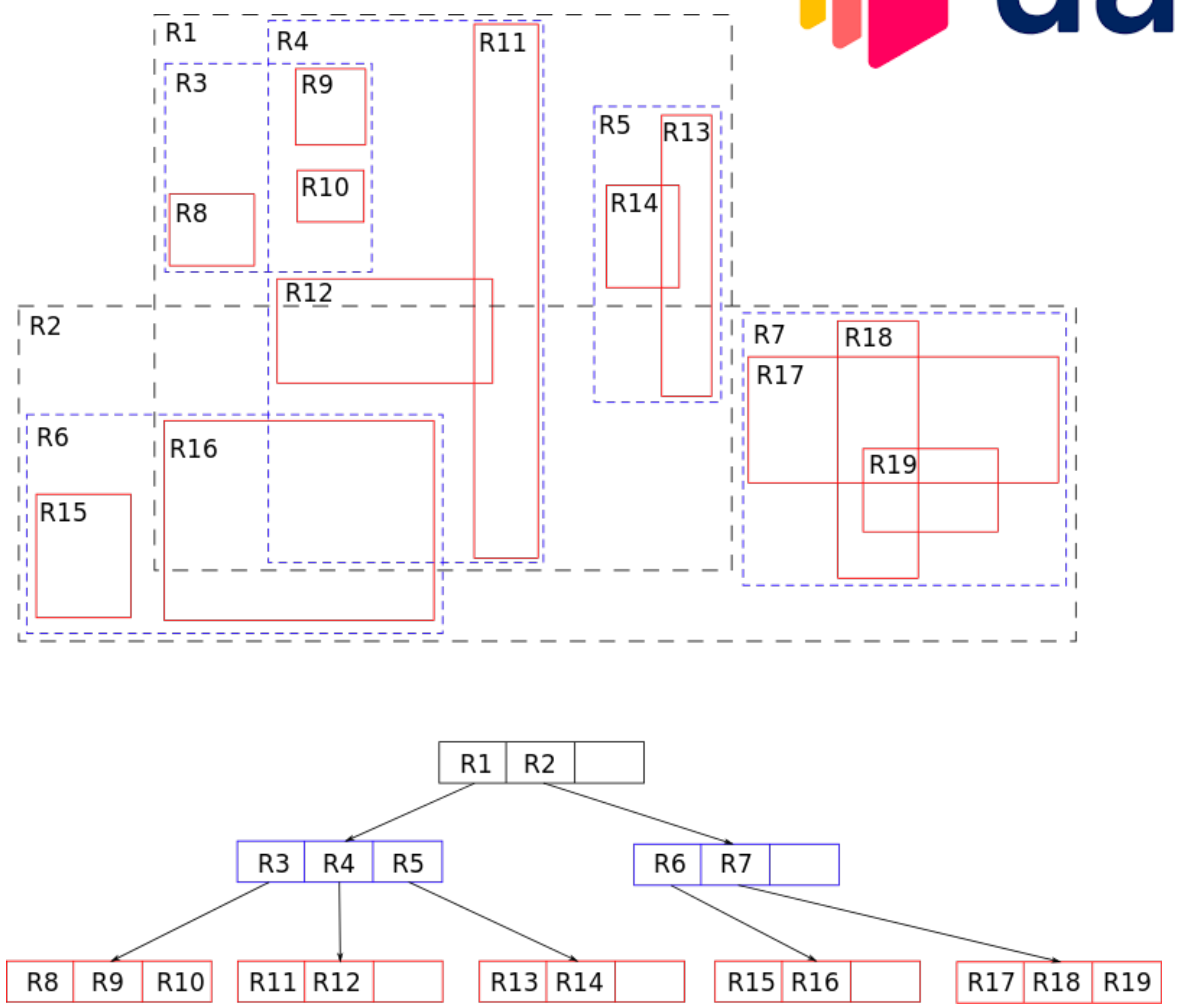
The choice of scale (MAUP) and classification scheme allows to create *arbitrary* interpretations



Aggregated data is simpler, but can be very biased

Next week: Big Spatial Data





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